Effective nonvanishing of K-trivial regular surfaces over imperfect fields

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Abstract

For any regular projective surface X over an arbitrary field k of characteristic $p > 0$ with $K_X \equiv 0$, we prove that

$$60K_X \sim 0.$$

This uniform bound is optimal. To show that the factor 5 is necessary, we construct an example in characteristic 5 for which K_X has exact order five.

If X is geometrically normal, then $12K_X \sim 0$. Our main focus is the geometrically non-normal case, where we obtain more precise bounds according to the Albanese morphism. If it is nontrivial, then $p = 2$ or 3, and we prove $12K_X \sim 0$ and $6K_X \sim 0$, respectively; both bounds are optimal. If it is trivial, then $K_X \sim 0$ for $p \geq 7$, while $5K_X \sim 0$, $3K_X \sim 0$, and $4K_X \sim 0$ for $p = 5, 3, 2$, respectively.

Contents

1 Introduction **1**
2 Preliminary **4**
3 The geometrically normal case **11**
4 Albanese fibrations and effective nonvanishing **12**
5 Torsion bounds with trivial Albanese variety **17**
6 Twisted covers and triviality in large characteristic **26**
7 Examples **29**

1 Introduction

For projective normal varieties X with mild singularities and $K_X \equiv 0$, which we call *K-trivial varieties*, the following questions arise naturally:

- (Nonvanishing) Is it true that $\kappa(X) \geq 0$? Equivalently, does $K_X \sim_{\mathbb{Q}} 0$ hold?
- (Effectivity) Does there exist a positive integer $N = N(d)$, depending only on the dimension d of X, such that $NK_X \sim 0$ holds for every such X?

For regular K-trivial surfaces over an algebraically closed field, it is well known that $12K_X \sim 0$; see [BdF07, BHPV04, BM77, BM76]. The same result holds over every perfect field by descent.

Over an algebraically closed field of characteristic 0, it is known that for terminal threefolds X with $K_X \equiv 0$, we have $2^5 \cdot 3^3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19K_X \sim 0$ (see [Bea83, Kaw86, Mor86, Ogu93]). One key idea in the proof is that when $q(X) := h^1(X, \mathcal{O}_X) > 0$, the Albanese morphism $X \to \mathrm{Alb}_X$ is an étale fiber bundle ([Kaw85]). This reduces the problem to a lower-dimensional case.

More generally, the Index Conjecture predicts that for complex projective klt pairs (X, D) with $K_X + D \sim_{\mathbb{Q}} 0$ and coefficients of D in a fixed DCC set $I \subset \mathbb{Q}$, the least positive integer m with $m(K_X + D) \sim 0$ is bounded in terms of $\dim X$ and I (see [Xu19] and the references therein). This is known in dimension at most three [Xu19, JL21] and in dimension four when $D \neq 0$ [Xu19].

Over an algebraically closed field of characteristic $p > 0$, not much is known. In [CWZ26b], we obtain a concrete description of K-trivial threefolds X with $q(X) > 0$. In the case $q = 2$, we proved $2^4 \cdot 3^3 K_X \sim 0$; for $q = 1$, we proved that the Albanese morphism $\mathrm{alb}_X \colon X \to \mathrm{Alb}_X$ is a fibration and $X \cong (E \times F)/G$, where E is an elliptic curve, F is a general fiber of alb_X and G is a finite group scheme acting diagonally on $E \times F$. Note that the surface F is not necessarily normal. Equivalently, the generic fiber X_η of alb_X, which is a regular surface over the function field $K(\mathrm{Alb}_X)$ with $K_{X_\eta} \equiv 0$, is not necessarily geometrically normal.

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More generally, fibrations whose generic fibers are regular K-trivial varieties arise naturally in birational geometry, including Iitaka fibrations defined by semi-ample canonical divisors and Albanese fibrations of K-trivial varieties. Studying geometrically non-normal regular K-trivial varieties over imperfect fields is therefore important for understanding such fibrations.

We study effective nonvanishing for K-trivial regular surfaces over imperfect fields. Tanaka proved nonvanishing in this setting, namely $K_X \sim_{\mathbb{Q}} 0$ ([Tan20, Theorem 1.1]). Our main result establishes effectivity as follows.

Theorem 1.1 (Propositions 3.2, 4.1, and 5.4; Corollary 6.2)**.** *Let X be a regular projective surface over a field k of characteristic $p > 0$ with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$. We have*

(A) *If X is geometrically normal, then $12K_X \sim 0$.*

(B) *If X is geometrically non-normal and the Albanese morphism $X \to \mathrm{Alb}_{X/k}$ is nontrivial, then $p = 2$ or 3; moreover, $12K_X \sim 0$ if $p = 2$, while $6K_X \sim 0$ if $p = 3$.*

(C) *If X is geometrically non-normal and the Albanese morphism of X is trivial, then*
- *if $p \geq 7$, then $K_X \sim 0$;*
- *if $p = 5$, then $5K_X \sim 0$;*
- *if $p = 3$, then $3K_X \sim 0$;*
- *if $p = 2$, then $4K_X \sim 0$.*

Corollary 1.2. *Let X be a regular projective surface over an arbitrary field k with $K_X \equiv 0$. Then*

$$60K_X \sim 0.$$

Moreover, if $\operatorname{char} k \neq 5$, then $12K_X \sim 0$.

The bound in Theorem 1.1(C) is sharp in characteristic 5.

Theorem 1.3 (Example 7.8)**.** *Let $k = \mathbf{F}_5(s_1, s_2)$, where s_1, s_2 are algebraically independent over $\mathbf{F}_5$. Then there exists a geometrically non-normal regular projective surface X over k with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$ such that*

$$\operatorname{ord}(K_X) = 5.$$

As a byproduct of the analysis of Case (B), we obtain the following description of the Albanese morphism of regular K-trivial surfaces. It can be viewed as a generalization of the (quasi-)bielliptic surface structure of [BM77, BM76].

Theorem 1.4 (Combine Propositions 3.1 and 4.1 with Lemma 2.25)**.** *Let X be a regular projective surface over a separably closed field k of characteristic $p > 0$, with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$. The Albanese morphism $\mathrm{alb}_{X/k} \colon X \to A := \mathrm{Alb}_{X/k}$ is a fibration. Moreover:*

(1) *If $\dim A = 2$, then $\mathrm{alb}_{X/k}$ is an isomorphism.*

(2) *If $\dim A = 1$, then $X \simeq (A \times_k C)/G$, where C is an Albanese fiber and $G \subset A$ is a finite subgroup scheme acting diagonally on the product.*

Our argument also gives the following result for numerically trivial line bundles on surfaces with nef anti-canonical divisor.

Theorem 1.5 (Propositions 5.4 and 5.5; Corollary 6.2)**.** *Let X be a regular projective surface over a field k of characteristic $p > 0$ with $H^0(X, \mathcal{O}_X) = k$. Assume that $-K_X$ is nef, $\dim \mathrm{Alb}_{X/k} = 0$, and X is geometrically non-normal. Then every numerically trivial line bundle L on X satisfies the following:*

(1) *If $p \geq 7$, then $L \sim 0$.*

(2) *If $p \in \{3, 5\}$, then $pL \sim 0$.*

(3) *If $p = 2$, then $2^2 L \sim 0$.*

Section 5 establishes the bounds $pL \sim 0$ for $p \geq 3$ and $4L \sim 0$ for $p = 2$. To obtain the stronger conclusion $L \sim 0$ when $p \geq 7$, we introduce the following construction.

The t-twisted μ_p-cover

We first recall the simple μ_n-cover (see for example [CDL25, §0.2]). Let X be a variety over a field k, let L be a line bundle on X, and let $n \geq 1$. A nonzero section $s \in H^0(X, L^{\otimes n})$ defines a finite flat cover of degree n

$$\pi_s \colon Y := \operatorname{Spec}_X \left(\bigoplus_{i=0}^{n-1} L^{-i} \right) \longrightarrow X,$$

where multiplication uses the map $L^{-n} \to \mathcal{O}_X$ given by evaluation on s. The grading defines a μ_n-action with quotient X. If s is nowhere vanishing, then π_s is a μ_n-torsor and $\pi_s^* L$ is trivial.

Now assume that $\operatorname{char} k = p > 0$ and fix a trivialization $\tau\colon L^{\otimes p} \xrightarrow{\sim} \mathcal{O}_X$, with $s := \tau^{-1}(1)$. The μ_p-cover associated to (L, s) need not be normal, even when X is regular. To obtain a regular cover, we introduce a transcendental twist.

For t transcendental over k, set $K = k(t)$ and $X_K = X \times_k K$. The simple cover associated to (L_K, ts_K) is called the *t-twisted μ_p-cover* and is denoted by $\pi\colon \widehat{Y} \to X_K$. Introducing t makes $\widehat{Y}$ regular whenever X is regular. This allows us to construct a geometrically non-reduced regular variety from some special p-torsion line bundle as follows.

Theorem 1.6 (Proposition 6.1)**.** *Let X be a regular projective variety over a field k of characteristic $p > 0$ with $H^0(X, \mathcal{O}_X) = k$. Assume that $X_{\bar{k}}$ is integral, and let $\nu\colon X^\nu_{\bar{k}} \to X_{\bar{k}}$ be its normalization. Let L be a line bundle on X with $\operatorname{ord}(L) = p$ and $\nu^* L_{\bar{k}} \simeq \mathcal{O}_{X^\nu_{\bar{k}}}$. Then the t-twisted μ_p-cover $\pi\colon \widehat{Y} \to X_K$ satisfies the following:*

(1) *$\widehat{Y}$ is a regular projective variety over K with $H^0(\widehat{Y}, \mathcal{O}_{\widehat{Y}}) = K$;*
(2) *$\omega_{\widehat{Y}} \simeq \pi^* \omega_{X_K}$ and $\pi^* L_K \simeq \mathcal{O}_{\widehat{Y}}$;*
(3) *$\widehat{Y}$ is geometrically non-reduced over K.*

By [WZ26, Theorem 1.2], a regular projective K-trivial surface over a field of characteristic $p \geq 7$ is geometrically integral. In Case (C) of Theorem 1.1 with $p \geq 7$, the proof of Proposition 5.4 shows that every numerically trivial line bundle L satisfies $L^{\otimes p} \simeq \mathcal{O}_X$ and becomes trivial on $X^\nu_{\bar{k}}$. If L were non-trivial, Theorem 1.6 would produce a regular geometrically non-reduced surface over K with numerically trivial canonical divisor, contradicting [WZ26, Theorem 1.2]. Thus $L \sim 0$, and in particular $K_X \sim 0$ (Corollary 6.2).

Strategy of the proof

We prove Case (A) by taking a geometric base change, resolving the resulting singularities, and then applying the classification of smooth surfaces and descent (Section 3).

In Cases (B) and (C), the Frobenius base change $X \times_k k^{1/p}$ is non-normal. Set $X_1 = (X \times_k k^{1/p})^\nu_{\mathrm{red}}$ to be the normalization, and let $\phi\colon X_1 \to X$ be the natural morphism. The canonical base-change formula (Theorem 2.10) gives

$$K_{X_1} + (p-1)\mathfrak{C} \sim \phi^* K_X \equiv 0, \qquad \mathfrak{C} > 0.$$

Thus K_{X_1} is not pseudo-effective. Resolving the singularities of X_1 and running the minimal model program gives the diagram

$$\begin{array}{ccccc} & & \widetilde{X}_1 & & \\ & \overset{\mu}{\swarrow} & & \overset{\epsilon}{\searrow} & \\ X & \xleftarrow{\phi} & X_1 & & Z \\ & & & & \downarrow{\scriptstyle g} \\ & & & & B\,. \end{array}$$

Here μ is the minimal resolution, ϵ is the MMP contraction, and g is a Mori fiber space. The pullback $\mu^*\phi^*\omega_X$ descends to a numerically trivial line bundle on the del Pezzo surface Z if $\dim B = 0$, or to a degree-zero line bundle on B if $\dim B = 1$. Since ϕ is purely inseparable of height one, a relation $m\phi^* K_X \sim 0$ for a positive integer m yields $pmK_X \sim 0$.

In Case (B), the nontrivial Albanese morphism forces B to be a curve of positive geometric genus. Let $f_1\colon X_1 \to B$ be the fibration induced by g. Restricting the canonical formula to a general fiber gives $p = 2$ or 3 and shows that $\mathfrak{C}$ contains an f_1-horizontal prime divisor C_1. A key step is to prove that $C_1^2 = 0$ and C_1 is semi-ample (Lemma 4.2). In higher dimensions, statements of this kind are proved via the positivity of direct images of pluricanonical sheaves [EP23, Eji23, CWZ26b]. But since our study is restricted to surfaces, we apply a more classical treatment following [Tan20, Proposition 3.1], which goes back to Mumford [Mum69, §2], see also [Mas93, Lemma, p. 682].

A sufficiently divisible multiple of C_1 therefore defines a second fibration $h_1\colon X_1 \to D_1$ transversal to f_1:

$$\begin{array}{ccc} X_1 & \xrightarrow{h_1} & D_1 \\ {\scriptstyle f_1}\downarrow & & \\ B\,. & & \end{array}$$

Here C_1 is vertical for h_1. On X, this yields a fibration transversal to the Albanese morphism, which is the analogue of the (quasi-)bielliptic structure in [BM77]. Adjunction along the fibers of h_1 shows that

X_1 is regular; calculations of normal bundles and descent then give $12K_X \sim 0$ for $p = 2$ and $6K_X \sim 0$ for $p = 3$ (Section 4).

In Case (C), either Z is del Pezzo or B has geometric genus zero. As explained before, we only need to show that $pK_X \sim 0$ in characteristic $p \geq 3$ and $4K_X \sim 0$ in characteristic 2.

For the del Pezzo case, every numerically trivial line bundle L satisfies $L \sim 0$ when $p \geq 5$, $3L \sim 0$ when $p = 3$, and $4L \sim 0$ when $p = 2$ [BT22, Theorem 4.1]. This is weaker than we need when $p = 2$ or 3, and we improve the bound under additional assumptions (Lemmas 5.1 and 5.2).

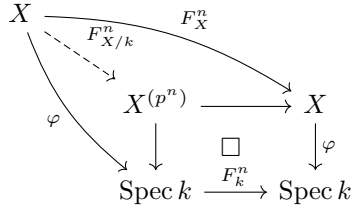
In the ruled case, the pullback $\mu^*\phi^*K_X$ descends to a degree-zero line bundle L_B on B. We analyze the horizontal components in the conductor divisor $\mathfrak{C}$ and use adjunction and positivity to bound the arithmetic genus of B by $p_a(B) \leq 1$. The line bundle L_B is trivial if $p_a(B) = 0$. When $p_a(B) = 1$, B is geometrically non-normal, and we can use the geometric singularities of B to obtain some properties on the geometric singularities of X, see Subcase 1(a) in the proof of Proposition 5.4 for example.

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AI Statement The authors wrote the first draft in 2025 and obtained a p-dependent bound on the torsion order of K_X. To improve the results, AI models were used — primarily GPT-5.6-Sol, GPT-6-astra and claude-opus-5. Specifically, AI provided the t-twisted μ_p-cover construction (§6), improved the result in Case (B) when $p = 3$ (Lemma 4.4), improved the result in Case (C) (part of §5) and constructed examples (§7).

2 Preliminary

2.1 Notation and conventions

- Throughout this paper, unless otherwise stated, k is a field of characteristic $p > 0$. We fix an algebraic closure $\bar{k}$ and write $k^{1/p^n} := \{a \in \bar{k} \mid a^{p^n} \in k\}$ for $n \geq 0$. The field k is called *F-finite* if $[k : k^p] < \infty$.
- By a k-scheme, we mean a separated scheme of finite type over k. By a *variety* over k we mean an integral k-scheme. By a *surface*, we mean a variety of dimension two.
- For a field extension ℓ/k and a k-scheme X, we write $X_\ell := X \times_k \ell$.
- For a scheme X, we denote by X_{red} the induced reduced scheme and by X^ν_{red} its normalization.
- A *fibration* is a projective surjective morphism $f\colon X \to Y$ of varieties such that $f_*\mathcal{O}_X = \mathcal{O}_Y$.
- For a line bundle L of finite order in $\mathrm{Pic}(X)$, we write $\mathrm{ord}(L)$ for its order; when ω_X is invertible, we write $\mathrm{ord}(K_X) := \mathrm{ord}(\omega_X)$.
- For an integral projective curve C over k, put $k_C := H^0(C, \mathcal{O}_C)$. We write
$$p_a(C) := \dim_{k_C} H^1(C, \mathcal{O}_C)$$
for its *arithmetic genus over the field of constants*. For a geometrically irreducible projective curve C, its *geometric genus* is the genus of $(C_{\bar{k}})^\nu_{\mathrm{red}}$ over $\bar{k}$.
- For a scheme X of characteristic $p > 0$, let $F_X\colon X \to X$ denote the absolute Frobenius, F_X^n its n-th iterate, and F_k^n the corresponding morphism of $\operatorname{Spec} k$.
- Let X be a k-scheme with structure morphism $\varphi\colon X \to \operatorname{Spec} k$, and set $X^{(p^n)} := X \times_{\operatorname{Spec} k, F_k^n} \operatorname{Spec} k$. The *$n$-th relative Frobenius morphism* $F^n_{X/k}\colon X \to X^{(p^n)}$ is the unique morphism making the following diagram commute:


- Let $\pi\colon X \to Y$ be a finite purely inseparable morphism. We say that π is of *height* n if there exists a morphism $\rho\colon Y \to X$ such that $\rho \circ \pi = F_X^n$.

2.2 Some basic results

Lemma 2.1. *Let $\pi\colon X \to Y$ be a finite surjective morphism of degree d between normal varieties.*

(1) *If D_Y is a Weil divisor on Y such that $\pi^* D_Y \sim 0$, then $dD_Y \sim 0$.*
(2) *If π is purely inseparable of height n and $\pi^* D_Y \sim 0$, then $p^n D_Y \sim 0$.*
(3) *If π is purely inseparable of height n and Y is regular, then $p^n D_X$ is Cartier for every Weil divisor D_X on X.*

Proof. See [CWZ26b, Lemma 6.8] for (1) and (2). To show (3), we factor the n-th iterate absolute Frobenius F_X^n into $X \xrightarrow{\pi} Y \xrightarrow{\rho} X$. Since Y is regular, the pullback divisor $\rho^* D_X$ is Cartier. Therefore, $p^n D_X = F_X^{n*} D_X = \pi^*(\rho^* D_X)$ is Cartier. □

Lemma 2.2. *Let k be an F-finite separably closed field of characteristic $p > 0$, and let T be a regular projective curve over k such that $H^0(T, \mathcal{O}_T) = k$ and T has geometric genus zero. Let M be a degree-zero line bundle on T.*

(1) *The line bundle M has p-power order.*
(2) *If $p_a(T) = 0$, then $M \sim 0$. If $p_a(T) = 1$, then $p \in \{2, 3\}$, the curve $T_1 := (T \times_k k^{1/p})^\nu_{\mathrm{red}}$ has arithmetic genus zero over its field of constants, and $pM \sim 0$.*

Proof. The assertions follow from [Tan21, Remark 5.2 and the proof of Theorem 11.8], Riemann–Roch, and Lemma 2.1(2). □

Lemma 2.3. *Let X be a scheme of finite type over k, and let $Z \subset X$ be a closed subscheme smooth over k. If Z is an effective Cartier divisor on X, then X is smooth over k on an open neighborhood of Z.*

Proof. Since Z is an effective Cartier divisor on X, the immersion $Z \hookrightarrow X$ is regular. The assertion follows from [SGA1, Exposé II, Corollary 4.16]. □

Lemma 2.4 (Riemann–Hurwitz, see [Liu02, Theorem 7.4.16])**.** *Let $\pi\colon B \to C$ be a finite* separable *morphism between regular projective curves over k. Then*

$$K_B \sim \pi^* K_C + \sum_x (e'_x - 1)[x], \qquad e'_x \geq e_x \geq 1,$$

where x runs over the closed points of B and e_x is the ramification index. Moreover, $e'_x = e_x$ precisely when π is tamely ramified at x, and $e'_x = 1$ precisely when π is étale at x.

2.3 Intersection numbers

Let X be a projective normal surface over k. Let D be a Cartier divisor on X, and let C be an integral curve on X. For a line bundle L on C, write $\deg_k L := \chi_k(C, L) - \chi_k(C, \mathcal{O}_C)$, where χ_k denotes the Euler characteristic computed over k. Thus $\deg_k L = [k_C : k] \deg_{k_C} L$, where $k_C = H^0(C, \mathcal{O}_C)$. The intersection number $(D \cdot C)_k$ of D on C is defined to be

$$(D \cdot C)_k := \deg_k\big(\mathcal{O}_X(D)|_C\big).$$

For a finite surjective morphism $\nu\colon C' \to C$ of integral projective curves over k, the projection formula gives

$$\deg_k\big(\nu^*(\mathcal{O}_X(D)|_C)\big) = [K(C') : K(C)](D \cdot C)_k.$$

In particular, for the normalization $\nu\colon C^\nu \to C$, we have $(D \cdot C)_k = \deg_k\big(\nu^*(\mathcal{O}_X(D)|_C)\big)$.

Let X be a geometrically irreducible variety over k, let ℓ/k be a finite field extension, and set $Y := (X \times_k \ell)^\nu_{\mathrm{red}}$. We define the *thickening exponent of this base change* by

$$\varepsilon(X; \ell/k) := \log_p \frac{[\ell : k]}{[K(Y) : K(X)]}.$$

It is a nonnegative integer, and it vanishes precisely when X_ℓ is reduced; cf. [Tan21, Theorem 7.3(1)–(2)].

Lemma 2.5. *Let X be a projective normal geometrically irreducible k-surface, D a Cartier divisor on X, and $C \subset X$ an integral curve. For a finite extension ℓ/k, let $f\colon Y := (X \times_k \ell)^\nu_{\mathrm{red}} \to X$ be the natural morphism. Then*

$$(D \cdot C)_k = p^{\varepsilon(X;\ell/k)}(f^* D \cdot f^* C)_\ell,$$

where $f^* C$ denotes the pullback as a Weil divisor.

Proof. Set $d = [K(Y) : K(X)]$. Since $f_* f^* C = dC$, the projection formula gives

$$[\ell : k](f^* D \cdot f^* C)_\ell = (D \cdot f_* f^* C)_k = d(D \cdot C)_k.$$

The assertion follows from $p^{\varepsilon(X;\ell/k)} = [\ell : k]/d$. □

Corollary 2.6. *Let X be a projective normal k-surface with $H^0(X, \mathcal{O}_X) = k$, let ℓ/k be a finite purely inseparable extension, and set $Y := (X_\ell)^\nu_{\mathrm{red}}$. For a $\mathbb{Q}$-Cartier integral curve $C_Y \subset Y$, its image $C_X \subset X$ is $\mathbb{Q}$-Cartier and*

$$C_Y^2 > 0 \quad (\textit{resp.} \ < 0, \ = 0) \quad \Longleftrightarrow \quad C_X^2 > 0 \quad (\textit{resp.} \ < 0, \ = 0).$$

2.4 Adjunction formula

Proposition 2.7 ([Kol13, Definition 2.34 and Proposition 2.35])**.** *Let X be a normal surface and $C \subset X$ a reduced curve with normalization $C^\nu \to C$. Assume that $K_X + C$ is $\mathbb{Q}$-Cartier. Then there exists an effective $\mathbb{Q}$-divisor Diff_{C^ν} on C^ν, called the different, such that*

$$(K_X + C)|_{C^\nu} \sim_{\mathbb{Q}} K_{C^\nu} + \mathrm{Diff}_{C^\nu}.$$

Moreover, $\mathrm{Diff}_{C^\nu} = 0$ if and only if C is regular and X is regular along C.

2.5 A semi-ampleness criterion

Proposition 2.8 (special case of [Tan20, Proposition 3.1])**.** *Let X be a projective normal surface over k. Let B be an integral curve on X, and assume that K_X and B are $\mathbb{Q}$-Cartier. Assume that*

(1) *$K_X \cdot B = B^2 = 0$;*
(2) *there exists a positive integer n such that nB is Cartier and $\mathcal{O}_X(nB)|_B \cong \mathcal{O}_B$.*

Then B is semi-ample.

2.6 Positivity

Lemma 2.9 (a special case of [Eji19, Corollary 4.6])**.** *Let k be an F-finite field of characteristic $p > 0$. Let $f \colon S \to C$ be a surjective morphism from a regular projective surface to a regular projective curve such that $H^0(C, \mathcal{O}_C) = k$, and let $\Delta \geq 0$ be an effective $\mathbb{Q}$-divisor whose Cartier index is not divisible by p. Assume that the geometric generic fiber $(S_{\bar{\eta}}, \Delta_{\bar{\eta}})$ is F-pure. Then the following statements hold.*

(1) *If $-(K_S + \Delta)$ is nef, then $\deg K_C \leq 0$.*
(2) *If $-(K_S + \Delta)$ is ample, then $\deg K_C < 0$.*
(3) *If $(S_{\bar{\eta}}, \Delta_{\bar{\eta}})$ is strongly F-regular and $-(K_S + \Delta)$ is nef and big, then $\deg K_C < 0$.*

2.7 Behavior of canonical divisors under purely inseparable field extensions

Let X be a normal variety over k such that k is algebraically closed in $K(X)$. Following [Tan21, Definition 5.1], the *Frobenius length of geometric non-normality* $\ell_F(X/k)$ is the least integer $n \geq 0$ such that

$$(X \times_k k^{1/p^n})^\nu_{\mathrm{red}}$$

is geometrically normal over k^{1/p^n}. This integer exists by [Tan21, Remark 5.2].

Theorem 2.10 ([Tan18a, Theorem 1.1] and [JW21, Theorem 1.1])**.** *Let X be a normal projective variety over a field k with $H^0(X, \mathcal{O}_X) = k$. Set either $\ell = k^{1/p}$ or $\ell = \bar{k}$. Let $Y := (X \times_k \ell)^\nu_{\mathrm{red}}$. Then there exists an effective $\mathbb{Z}$-Weil divisor $\mathfrak{C} = \mathfrak{M} + \mathfrak{F}$ such that*

$$-K_{Y/X} \sim (p-1)\mathfrak{C}, \tag{2.1}$$

where

- *$\mathfrak{C} = 0$ if and only if X is geometrically normal;*
- *$\mathfrak{F}$ is supported on the conductor divisor of the normalization map $Y \to (X \times_k \ell)_{\mathrm{red}}$; and*
- *$\mathfrak{M}$ is a movable divisor, and $\mathfrak{M} \neq 0$ if and only if X is geometrically non-reduced.*

The original statement of [JW21, Theorem 1.1] assumes that k is a function field, but this assumption is unnecessary as noted in [DW22, page 3917].

We recall how the divisor $\mathfrak{C} = \mathfrak{M} + \mathfrak{F}$ is constructed in [JW21, §3.2] in terms of foliations when $\ell = k^{1/p}$ and k is F-finite. Let $r := \dim_\ell \mathrm{Der}_k(\ell)$. Let $\phi \colon Y \to X$ be the induced morphism, and let

$$\mathscr{G} := T_{Y/X} = \mathcal{H}om_{\mathcal{O}_Y}(\Omega_{Y/X}, \mathcal{O}_Y)$$

be the foliation associated with ϕ. Restricting local derivations in $\mathscr{G}$ to the subfield $\ell \subset K(Y)$ gives a natural inclusion

$$\mathscr{G} \hookrightarrow \mathrm{Der}_k(\ell) \otimes_\ell \mathcal{O}_Y \simeq \mathcal{O}_Y^{\oplus r}.$$

Let $\mathscr{G}'$ be the saturation of $\mathscr{G}$ in this trivial bundle, and denote the inclusion by $\iota\colon \mathscr{G} \hookrightarrow \mathscr{G}'$. Then we have the exact sequences

$$0 \longrightarrow \mathscr{G} \xrightarrow{\iota} \mathscr{G}' \longrightarrow \mathrm{coker}(\iota) \longrightarrow 0 \tag{2.2}$$

and

$$0 \longrightarrow \mathscr{G}' \longrightarrow \mathrm{Der}_k(\ell) \otimes_\ell \mathcal{O}_Y \longrightarrow \mathscr{Q} \longrightarrow 0, \tag{2.3}$$

where $\mathscr{Q}$ is torsion-free by definition of saturation. The zero divisor of $\det(\iota)$ is the fixed divisor $\mathfrak{F}$ from Theorem 2.10; see [JW21, Definition 3.2]. If $X \times_k \ell$ is reduced (equivalently, $\mathfrak{M} = 0$), then $\mathscr{G}' = \mathrm{Der}_k(\ell) \otimes_\ell \mathcal{O}_Y \simeq \mathcal{O}_Y^{\oplus r}$, so $\mathfrak{F}$ is the zero divisor of det of the inclusion $\mathscr{G} \hookrightarrow \mathcal{O}_Y^{\oplus r}$; see [JW21, Remark 3.3].

To describe $\mathfrak{M}$, we keep track of how the fiber of $\mathscr{G}'$ inside $V := \mathrm{Der}_k(\ell)$ varies with the point of Y. Put $q := \mathrm{rank}\,\mathscr{G}$ and $s := r - q = \mathrm{rank}\,\mathscr{Q}$. At a general point $y \in Y$, the fiber of $\mathscr{G}'$ is a q-dimensional subspace of $V \otimes_\ell \kappa(y)$; sending y to this subspace and composing with the Plücker embedding gives a rational map (cf. [JW21, §2.3])

$$\gamma_0\colon Y \dashrightarrow \mathrm{Gr}(q, V) \hookrightarrow \mathbb{P}(\wedge^s V). \tag{2.4}$$

We call γ_0 the Gauss map associated with $\mathscr{G}$, using the same notation for its composition with the Plücker embedding. Its domain is the open locus $U \subseteq Y$ where $\mathscr{G}'$ is a subbundle of $V \otimes_\ell \mathcal{O}_Y$, i.e. where $\mathscr{Q}$ is locally free, whose complement has codimension at least 2 since Y is normal and $\mathscr{Q}$ is torsion-free. Taking s-th exterior powers of $V \otimes_\ell \mathcal{O}_Y \twoheadrightarrow \mathscr{Q}$ gives a map $\wedge^s V \otimes_\ell \mathcal{O}_Y \to \det \mathscr{Q}$, which is surjective on U, where it is the pullback by γ_0 of $\wedge^s V \otimes \mathcal{O} \twoheadrightarrow \mathcal{O}(1)$. Hence the sections of $\det \mathscr{Q}$ coming from $\wedge^s V$ span the linear system $\mathcal{M} = \gamma_0^* |\mathcal{O}(1)|$, whose base locus is exactly $Y \setminus U$, of codimension at least 2; in particular, $\mathcal{M}$ is movable. Moreover, $\det \mathscr{G}' \simeq (\det \mathscr{Q})^\vee$ on U, hence on Y by reflexivity. By [JW21, Definition 5.1 and Lemma 5.2], the divisor $\mathfrak{M}$ of Theorem 2.10 is a member of $\mathcal{M}$, so

$$\mathcal{M} \subseteq |\mathfrak{M}|, \qquad \mathcal{O}_Y(\mathfrak{M}) \simeq (\det \mathscr{G}')^\vee \simeq \det \mathscr{Q}.$$

In particular, γ_0 is a morphism along any curve $C \subseteq Y$ that is disjoint from some member of $\mathcal{M}$. Since $\mathfrak{F}$ is the zero divisor of $\det(\iota)$, the isomorphism $\mathcal{O}_Y(\mathfrak{M}) \simeq (\det \mathscr{G}')^\vee$ gives

$$c_1(\mathscr{G}') = -\mathfrak{M}, \qquad c_1(\mathscr{G}) = c_1(\mathscr{G}') - \mathfrak{F} = -\mathfrak{C}. \tag{2.5}$$

Lemma 2.11. *In the above setting with $\ell = k^{1/p}$, let $a \in \ell \setminus k$, and let D be an effective Weil divisor on Y. Assume that $v_P(\partial(a)) \geq \mathrm{mult}_P D$ for every prime divisor $P \subseteq \mathrm{Supp}\, D$ and every ∂ in the stalk of $\mathscr{G}$ at the generic point of P. Then $\mathfrak{C} \sim D + E$ for some effective divisor E on Y.*

Proof. Since $a \in \ell \subseteq H^0(Y, \mathcal{O}_Y)$ and local sections of $\mathscr{G}$ are derivations of $\mathcal{O}_Y$, the function $\partial(a)$ is regular for every local section ∂ of $\mathscr{G}$. The hypothesis, normality of Y, and reflexivity of $\mathcal{O}_Y(-D)$ therefore give a map [SP, Tag 0AVC]

$$\mathrm{ev}_a\colon \mathscr{G} \longrightarrow \mathcal{O}_Y(-D), \qquad \partial \longmapsto \partial(a).$$

This map is nonzero since $a \notin K(X)$. Extend a to a p-basis $a_1 = a, a_2, \ldots, a_r$ of ℓ/k. For $2 \leq i \leq r$, define

$$\mathrm{ev}_i\colon \mathscr{G} \longrightarrow \mathcal{O}_Y, \qquad \partial \longmapsto \partial(a_i).$$

The inclusion $\mathscr{G} \hookrightarrow \mathrm{Der}_k(\ell) \otimes_\ell \mathcal{O}_Y$ implies that $\mathrm{ev}_a, \mathrm{ev}_2, \ldots, \mathrm{ev}_r$ span the dual of $\mathscr{G}$ at the generic point. After renumbering $a_2, \ldots, a_r$, we may therefore assume that $\mathrm{ev}_a, \mathrm{ev}_2, \ldots, \mathrm{ev}_q$ are linearly independent there. Taking the exterior product $\mathrm{ev}_a \wedge \mathrm{ev}_2 \wedge \cdots \wedge \mathrm{ev}_q$ and reflexive hulls, and using (2.5), gives a nonzero map

$$\mathcal{O}_Y(-\mathfrak{C}) \simeq \det \mathscr{G} \longrightarrow \mathcal{O}_Y(-D).$$

Hence $\mathfrak{C} \sim D + E$ for some effective divisor E. □

2.8 Del Pezzo surfaces and Mori fiber spaces

Definition 2.12. Let X be a regular projective k-surface with $H^0(X, \mathcal{O}_X) = k$. We say that X is a *del Pezzo surface* if $-K_X$ is ample. We say that a fibration $f\colon X \to B$ from X is a *Mori fiber space* if $\dim B < \dim X$, $\rho(X/B) = 1$ and $-K_X$ is relatively ample over B.

In this paper, we need the following results.

Proposition 2.13 ([BT22, Theorem 4.1])**.** *Let X be a normal projective surface over k such that X has at most canonical singularity and $-K_X$ is nef and big. Let $L \in \mathrm{Pic}(X)$ such that $L \equiv 0$. Then:*

(1) *if* $p \geq 5$, *then* $L \sim 0$;
(2) *if* $p = 3$, *then* $3L \sim 0$;
(3) *if* $p = 2$, *then* $2^2 L \sim 0$.

Proposition 2.14 (see [BT22, Proposition 2.18])**.** *Let* $f\colon X \to B$ *be a* K_X*-Mori fibre space from a regular projective surface to a curve with* $H^0(B, \mathcal{O}_B) = k$*. Let* $b \in B$ *be a (not necessarily closed) point. Then the fibre* X_b *is an integral* $\kappa(b)$*-conic with* $\kappa(b) = H^0(X_b, \mathcal{O}_{X_b})$*.*

Definition 2.15. Let K be a field of characteristic two. A conic in $\mathbb{P}^2_K$ is *quasilinear* if it is defined by an equation $sx^2 + ty^2 + rz^2 = 0$ with $s, t, r \in K$.

By [Tan21, Theorem 9.10 (5), (6)], a regular conic F over a field K with $H^0(F, \mathcal{O}_F) = K$ is either smooth over K, or $\operatorname{char} K = 2$ and F is quasilinear.

2.9 Minimal model program of surfaces

Definition 2.16. Let X be a regular projective surface over k. We call an integral curve $C \subset X$ a (-1)*-curve* if

$$(K_X \cdot C)_{k_C} = (C)^2_{k_C} = -1$$

where $k_C = H^0(C, \mathcal{O}_C)$. Note that, for such a curve C, we have $H^1(C, \mathcal{O}_C) = 0$.

Theorem 2.17 (Castelnuovo's contraction criterion (special case of [Lip69, Theorem 27.1]))**.** *Let* $C \subset X$ *be a* (-1)*-curve on a regular projective surface* X*. Then there exists a birational contraction* $f\colon X \to Z$ *to a regular surface* Z *such that* C *is the exceptional curve.*

Theorem 2.18 (minimal model program ([Sha66, page 97], [Tan18b, page 5]))**.** *Let* X *be a regular projective surface over* k*. Then every sequence of contractions of* (-1)*-curves terminates at a regular surface* Z *such that either* K_Z *is nef or* Z *admits a Mori fiber space structure.*

Lemma 2.19. *Let* $\pi\colon X \to Y$ *be a purely inseparable morphism of height* n *between normal surfaces, and assume that* X *has rational singularities. Then every exceptional curve* E *of a resolution* $\mu\colon \widetilde{Y} \to Y$ *admits a purely inseparable cover* $C \to E$ *of height* n *such that* $H^1(C, \mathcal{O}_C) = 0$*.*

Proof. Let W be the normalization of $\widetilde{Y}$ in $K(X)$. The induced morphisms fit into a commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{\rho} & \widetilde{Y} \\ {\scriptstyle\nu}\downarrow & & \downarrow{\scriptstyle\mu} \\ X & \xrightarrow{\pi} & Y\,, \end{array}$$

where ρ is purely inseparable of height n and ν is proper birational. Let $y = \mu(E)$, let x be the unique point of X over y, and set $F = \nu^{-1}(x)$, $C = \left(\rho^{-1}(E)\right)_{\mathrm{red}}$. Thus, C is an irreducible component of F, and $C \to E$ is a purely inseparable cover of height n. Since X has rational singularities, $H^1(F, \mathcal{O}_F) = 0$ [Lip69, Proposition 1.2]. Since F is one-dimensional, the surjection $\mathcal{O}_F \twoheadrightarrow \mathcal{O}_C$ induces a surjection on H^1, so $H^1(C, \mathcal{O}_C) = 0$. □

Construction 2.20 (MMP of the base change)**.** Let X be a regular projective k-surface with $H^0(X, \mathcal{O}_X) = k$, let ℓ be $k^{1/p}$ or $\bar{k}$, and let $\phi\colon Y := (X \times_k \ell)^{\nu}_{\mathrm{red}} \to X$ be the natural morphism. Let $\mu\colon \widetilde{Y} \to Y$ be a minimal resolution. By Theorem 2.18, any sequence

$$\widetilde{Y} =: Z_0 \overset{\epsilon_0}{\to} Z_1 \overset{\epsilon_1}{\to} Z_2 \to \cdots \to Z_n$$

of contractions of (-1)-curves terminates at $Z := Z_n$, where

- either K_Z is nef;
- or Z admits a Mori fiber space structure $g\colon Z \to B$.

Let $\epsilon\colon \widetilde{Y} \to Z$ be the composite contraction and, in the second case, set $f := g \circ \epsilon$. In the Mori fiber space case, the morphisms fit into the diagram

$$\begin{array}{ccccc} & & \widetilde{Y} & \xrightarrow{\epsilon} & Z \\ & & {\scriptstyle\mu}\downarrow & \searrow^{f} & \downarrow{\scriptstyle g} \\ X & \xleftarrow{\phi} & Y & & B \end{array} \tag{2.6}$$

By Theorem 2.10, we have

$$K_Y + (p-1)\mathfrak{C} = \phi^* K_X,$$

where $\mathfrak{C} = 0$ if and only if X is geometrically normal. Then

$$\mu^*\phi^*K_X = K_{\widetilde{Y}} + (p-1)\widetilde{\mathfrak{C}} + \sum m_i E_i, \tag{2.7}$$

where $\widetilde{\mathfrak{C}}$ is the strict transform of $\mathfrak{C}$ and the E_i are the μ-exceptional curves. The coefficients m_i are nonnegative integers: nonnegativity follows from the minimality of μ and the negativity lemma, while integrality follows because K_X is Cartier.

Lemma 2.21. *Consider a commutative diagram*

$$\begin{array}{ccc} Y & \xrightarrow{\epsilon} & Z \\ & \searrow^{f} & \downarrow g \\ & & B, \end{array}$$

where Y and Z are regular projective k-surfaces, ϵ is a composition of contractions of (-1)-curves, g is a Mori fiber space, and $f = g \circ \epsilon$. Let $M \in \mathrm{Pic}(Y)$ satisfy $M \equiv 0$. Then the following statements hold.

(1) *If $\dim B = 0$, then there exists $M_Z \in \mathrm{Pic}(Z)$ such that $M \cong \epsilon^* M_Z$.*
(2) *If $\dim B = 1$, then there exists $M_B \in \mathrm{Pic}(B)$ such that $M \cong f^* M_B$.*

Proof. Both assertions follow by applying [Tan18b, Theorem 4.4(3)] successively to the contractions in ϵ and, when $\dim B = 1$, to g. □

2.10 Albanese variety over an arbitrary field

The existence of the Albanese morphism over an algebraically closed field was established by Serre [Ser60], and later by Grothendieck using a more advanced approach (cf. [FGI+05, §9.5]). This was extended to geometrically integral varieties over arbitrary fields in [Wit08, Appendix A]; see also [Con17] and [Lau22, §1]. More recently, [LS24], [Sch24] and [ACMV25, §2.2] established a more general result that applies to geometrically non-reduced varieties.

Proposition 2.22 (special case of [LS24, Theorem 10.2] or [Sch24, Theorem 5.4])**.** *Let X be a proper variety over k with $H^0(X, \mathcal{O}_X) = k$. Then there exists a* para-abelian variety $\mathrm{Alb}_{X/k}$ *over k (by which we mean that $\mathrm{Alb}_{X/k} \times_k \bar{k}$ admits an abelian variety structure) and a morphism $\mathrm{alb}_{X/k} \colon X \to \mathrm{Alb}_{X/k}$ verifying the following universal property:*

- *for every para-abelian variety A and every morphism $g \colon X \to A$, there exists a unique morphism $\phi \colon \mathrm{Alb}_{X/k} \to A$ such that $g = \phi \circ \mathrm{alb}_{X/k}$.*

Every para-abelian variety carries a canonical simply transitive action of an abelian variety. If k is separably closed, then $\mathrm{Alb}_{X/k}$ has a k-point, hence it is an abelian variety.

Lemma 2.23. *Let X be a proper variety over k with $H^0(X, \mathcal{O}_X) = k$.*

(1) *If X is geometrically connected and geometrically reduced, then for any field extension $k \subset k'$, there is a natural isomorphism $\mathrm{Alb}_{X_{k'}/k'} \xrightarrow{\sim} \mathrm{Alb}_{X/k} \otimes_k k'$.*
(2) *Let $Y := (X \times_k k^{1/p})^\nu_{\mathrm{red}}$. Then $\dim(\mathrm{Alb}_{Y/k^{1/p}}) = \dim(\mathrm{Alb}_{X/k})$.*
(3) *Let $Y := (X \times_k \bar{k})^\nu_{\mathrm{red}}$. Then $\dim \mathrm{Alb}_{Y/\bar{k}} = \dim \mathrm{Alb}_{X/k}$.*

Proof. (1) See [ACMV25, Theorem 3.2].

(2) Set $k_1 = k^{1/p}$. We have $H^0(Y, \mathcal{O}_Y) = k_1$, and thus the Albanese variety Alb_{Y/k_1} exists. Consider the following commutative diagram

$$\begin{array}{ccccccccc} & & & & & & F_X & & \\ & & \overset{g}{\nearrow} & & \overset{\nu}{\nearrow} & & \nearrow & & \\ X^{\frac{1}{p}} & & \longrightarrow & Y & \longrightarrow & X_{k_1} & \longrightarrow & X & \\ \downarrow{\scriptstyle \mathrm{alb}_{X^{\frac{1}{p}}/k_1}} & & & \downarrow{\scriptstyle \mathrm{alb}_{Y/k_1}} & & \downarrow{\scriptstyle \mathrm{alb}_{X_{k_1}/k_1}} & \searrow^{(\mathrm{alb}_{X/k})_{k_1}} & \downarrow & \\ \mathrm{Alb}_{X^{\frac{1}{p}}/k_1} & & \xrightarrow{g'} & \mathrm{Alb}_{Y/k_1} & \xrightarrow{\nu'} & \mathrm{Alb}_{X_{k_1}/k_1} & \xrightarrow{u} & \mathrm{Alb}_{X/k} \otimes_k k_1 & \longrightarrow \mathrm{Alb}_{X/k} \end{array}$$

where $X^{\frac{1}{p}}$ denotes X regarded as a k_1-scheme through the isomorphism $k_1 \to k$, $a \mapsto a^p$, the morphisms g' and ν' are induced by the functoriality of Alb ([LS24, Corollary 10.4]), and u is induced by the universal property of $\mathrm{Alb}_{X_{k_1}/k_1}$. By the uniqueness of Alb ([LS24, Corollary 10.4]), the composition of the second

line in the above diagram is the absolute Frobenius of $\mathrm{Alb}_{X/k}$, which is a homeomorphism. Hence g' has finite fibers. Since g is surjective and $\mathrm{alb}_{Y/k_1} \circ g = g' \circ \mathrm{alb}_{X^{\frac{1}{p}}/k_1}$, the image of g' contains $\mathrm{alb}_{Y/k_1}(Y)$, which generates Alb_{Y/k_1}, so g' is surjective. Therefore $\dim \mathrm{Alb}_{Y/k_1} = \dim \mathrm{Alb}_{X^{\frac{1}{p}}/k_1} = \dim \mathrm{Alb}_{X/k}$.

(3) Since there exists a finite purely inseparable field extension $k \subset k' \subseteq \bar{k}$ such that $(X \times_k k')^{\nu}_{\mathrm{red}}$ is geometrically integral, the result of (3) follows from (1) and (2). □

Over an algebraically closed field, the following result was proved for smooth varieties in [HPZ19, Theorem 0.3] and extended to normal varieties in [EP23, Proposition 2.3].

Proposition 2.24. *Let X be a normal projective variety over a field k such that $H^0(X, \mathcal{O}_X) = k$ and $-K_X$ is nef. If X is of maximal Albanese dimension (i.e., $\dim X = \dim \mathrm{alb}_{X/k}(X)$), then $\mathrm{alb}_{X/k} \colon X \to \mathrm{Alb}_{X/k}$ is an isomorphism.*

Proof. Let $Y := (X \times_k \bar{k})^{\nu}_{\mathrm{red}}$ and let $\phi \colon Y \to X$ be the induced morphism. Then $K_Y + (p-1)\mathfrak{C} = \phi^* K_X$ for some effective divisor $\mathfrak{C}$, where $\mathfrak{C} = 0$ if and only if X is geometrically normal. By Lemma 2.23, Y is of maximal Albanese dimension. Thus $\kappa(Y, K_Y) \geq 0$ (see [CWZ26a, Proposition 2.9 (1)]). Since $-K_X$ is nef, this implies that $\mathfrak{C} = 0$ and $K_Y \equiv \phi^* K_X \equiv 0$. Consequently, X is geometrically normal and $Y \cong X \times_k \bar{k}$. Since Y is of maximal Albanese dimension and $K_Y \equiv 0$, it follows from [EP23, Proposition 2.3] that Y is isomorphic to an abelian variety. Thus, X is a para-abelian variety which is equivalent to the assertion. □

Lemma 2.25. *Let X be a regular projective surface over a field k with $H^0(X, \mathcal{O}_X) = k$. Suppose that $E := \mathrm{Alb}_{X/k}$ is an elliptic curve, and write $\mathrm{alb} := \mathrm{alb}_{X/k}$. Let $h \colon X \to B$ be a fibration onto a regular projective curve whose generic fiber Γ satisfies $p_a(\Gamma) = 1$. Assume that $K_{X/B}$ is h-nef and that* alb *does not factor through h.*

$$\begin{array}{ccc} X & \xrightarrow{h} & B \\ \downarrow{\scriptstyle \mathrm{alb}} & & \\ E & & \end{array}$$

Then the following statements hold.

(1) *The Albanese morphism* alb *is a fibration.*

(2) *Choose an origin 0 on E and set $C := \mathrm{alb}^{-1}(0)$. There exist an integer $n \geq 1$ and an action of $E[n]$ on C such that*

$$X \simeq (E \times_k C)/E[n],$$

where $E[n]$ acts on E by translations and on $E \times_k C$ diagonally.

Proof. We adapt the argument of [BM77, §3, pp. 34–35].

Let $K = K(B)$, $E_K := E \times_k \operatorname{Spec} K$, $J = \mathrm{Pic}^0_{\Gamma/K}$, and $\alpha := \mathrm{alb}|_\Gamma \colon \Gamma \to E_K$. The morphism α is dominant, since otherwise alb would factor through h. The generic fiber Γ is regular, has $H^0(\Gamma, \mathcal{O}_\Gamma) = K$ and $K_\Gamma \sim 0$, and dominates the elliptic curve E_K. Proposition 2.24 therefore shows that Γ is a smooth genus-one curve, with its natural simply transitive J-action $\mu \colon J \times_K \Gamma \longrightarrow \Gamma$. After a finite Galois extension L/K, choose $x_0 \in \Gamma(L)$. The identification $J_L \simeq \Gamma_L$ given by $j \mapsto \mu(j, x_0)$ defines a morphism

$$u_L \colon J_L \longrightarrow E_L, \qquad u_L(j) := \alpha_L(\mu(j, x_0)) - \alpha_L(x_0).$$

Since $u_L(0) = 0$, u_L is a homomorphism of abelian varieties. Replacing x_0 by $\mu(j_0, x_0)$ changes the right-hand side to $u_L(j + j_0) - u_L(j_0) = u_L(j)$. Thus u_L is independent of x_0, hence Galois-invariant, and descends to a homomorphism

$$u \colon J \to E_K.$$

Since α is dominant, u is nonzero and hence an isogeny. The construction gives

$$\alpha(\mu(j, x)) = u(j) + \alpha(x)$$

for every K-scheme S and all $j \in J(S)$ and $x \in \Gamma(S)$. For $n := \deg u$, the dual isogeny $v \colon E_K \to J$ satisfies $uv = [n]$. Define the E_K-action on Γ by

$$t \cdot x := \mu(v(t), x). \tag{2.8}$$

Then

$$\alpha(t \cdot x) = \alpha(\mu(v(t), x)) = u(v(t)) + \alpha(x) = \alpha(x) + [n]t. \tag{2.9}$$

Since $K_{X/B}$ is h-nef, [Kol25, Theorem 7] extends the E_K-action (2.8) to an action of $E \times_k B$ on X over B, equivalently an action

$$E \times_k X \longrightarrow X, \quad (t, x) \longmapsto t \cdot x, \quad \text{satisfying } h(t \cdot x) = h(x).$$

Equation (2.9) then extends to

$$\operatorname{alb}(t \cdot x) = \operatorname{alb}(x) + [n]t, \tag{2.10}$$

since each side is a morphism $E \times_k X \to E$ and they are equal on a dense subset.

Set $C := \operatorname{alb}^{-1}(0)$ and form the Cartesian diagram

$$\begin{array}{ccc} X' := E \times_{[n],E,\operatorname{alb}} X & \xrightarrow{\pi} & X \\ {\scriptstyle a'}\downarrow & & \downarrow{\scriptstyle \operatorname{alb}} \\ E & \xrightarrow[[n]]{} & E. \end{array}$$

Equation (2.10) gives an isomorphism

$$E \times_k C \xrightarrow{\sim} X', \qquad (t, z) \longmapsto (t, t \cdot z), \tag{2.11}$$

with inverse $(t, x) \mapsto (t, (-t) \cdot x)$. To see that this inverse is well-defined, note that $(t, x) \in X'$ satisfies $\operatorname{alb}(x) = [n]t$, so (2.10) gives

$$\operatorname{alb}((-t) \cdot x) = \operatorname{alb}(x) - [n]t = 0,$$

and hence $(-t) \cdot x \in C$. Write $X \xrightarrow{f} B_0 \xrightarrow{g} E$ for the Stein factorization of alb. Then B_0 is a regular curve with $H^0(B_0, \mathcal{O}_{B_0}) = k$, and g is finite flat. By flat base change, the Stein factorization of a' is

$$X' \longrightarrow B' := E \times_{[n],E,g} B_0 \longrightarrow E.$$

Via (2.11), a' is the projection $E \times_k C \to E$, whose Stein factorization is

$$E \times_k C \longrightarrow E \times_k T \longrightarrow E, \qquad T := g^{-1}(0) \simeq \operatorname{Spec} H^0(C, \mathcal{O}_C).$$

Hence $B' \simeq E \times_k T$. The scheme T is a zero-dimensional Cartier divisor on the regular curve B_0, hence it is Gorenstein. Its dualizing sheaf is therefore invertible, and hence trivial because T is finite over k. Hence $\omega_{B'/k} \simeq \mathcal{O}_{B'}$. Let $\pi_0 \colon B' \to B_0$ be the projection. The isogeny $[n] \colon E \to E$ has trivial relative dualizing sheaf, so flat base change and transitivity of duality give

$$\pi_0^* \omega_{B_0/k} \simeq \omega_{B'/k} \simeq \mathcal{O}_{B'}.$$

Taking the norm along the finite locally free morphism π_0, of degree n^2, yields $\omega_{B_0/k}^{\otimes n^2} \simeq \mathcal{O}_{B_0}$, hence $p_a(B_0) = 1$. Since $g \colon B_0 \to E$ is finite and dominant, Proposition 2.24 shows that B_0 is a smooth curve of genus one. The Albanese universal property shows that g is an isomorphism, so alb is a fibration.

Finally, π is an $E[n]$-torsor, being the base change of $[n]$. The torsor action on X' is given by $s \cdot (t, x) = (t + s, x)$ for $s \in E[n]$. Under (2.11), its action becomes

$$s \cdot (t, z) = (t + s, (-s) \cdot z), \qquad s \in E[n].$$

This is a free diagonal action with quotient X. □

3 The geometrically normal case

Proposition 3.1. *Let X be a regular projective surface over a field k with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$. Assume that X is geometrically normal, and let $\mu \colon \widetilde{Y} \to Y := X_{\bar{k}}$ be the minimal resolution. Then either $\widetilde{Y}$ is rational, or μ is crepant.*

Proof. We may assume that k is separably closed. Write

$$\mu^* K_Y = K_{\widetilde{Y}} + \sum_i m_i E_i \equiv 0, \quad \text{where the } E_i \text{ are the } \mu\text{-exceptional curves and } m_i \geq 0.$$

Recall that μ is *crepant* if $K_{\widetilde{Y}} = \mu^* K_Y$, that is, if $m_i = 0$ for all i. Suppose that μ is not crepant, so that some $m_i > 0$. Then $K_{\widetilde{Y}}$ is not pseudo-effective, so the MMP ends with a Mori fiber space $g\colon Z \to B$. The resulting morphisms fit into the diagram as in Construction 2.20:

$$\begin{array}{ccccc} & & \widetilde{Y} & \xrightarrow{\epsilon} & Z \\ & & \downarrow{\scriptstyle \mu} & \searrow{\scriptstyle f} & \downarrow{\scriptstyle g} \\ X & \xleftarrow{\phi} & Y & & B\,. \end{array}$$

If $\dim B = 0$, then Z is a smooth del Pezzo surface and hence rational, so $\widetilde{Y}$ is rational too. Suppose now that $\dim B = 1$. After descending the resolution to k^{1/p^n} for some n, we apply Lemma 2.19 to the relative Frobenius $X^{1/p^n} \to X_{k^{1/p^n}}$ to conclude that every E_i is rational. If $p_a(B) > 0$, this implies that every E_i is vertical over B, so pushing forward $K_{\widetilde{Y}} + \sum_i m_i E_i \equiv 0$ along ϵ gives $K_Z + V \equiv 0$ for some vertical divisor V, which contradicts the relative ampleness of $-K_Z$ over B. Therefore $p_a(B) = 0$, so $\widetilde{Y}$ is rational. □

Proposition 3.2. *Let X be a regular projective surface over k with $H^0(X, \mathcal{O}_X) = k$ and $K_X \equiv 0$. If X is geometrically normal, then $12K_X \sim 0$.*

Proof. By assumption, the base change $Y := X \times_k \bar{k}$ is a normal surface. Since the pullback $\mathrm{Pic}(X) \to \mathrm{Pic}(Y)$ is injective by [SP, Tag 0CC5], it is enough to show that $12K_Y \sim 0$. Let $\mu\colon \widetilde{Y} \to Y$ be the minimal resolution. By Proposition 3.1, either μ is crepant or $\widetilde{Y}$ is rational. If μ is crepant, then $K_{\widetilde{Y}} \equiv 0$, and it follows that $12K_{\widetilde{Y}} \sim 0$ ([BM77, BM76]). In turn, we have $12K_Y \sim 0$. If $\widetilde{Y}$ is rational, then $\mu^* K_Y \sim 0$, since a smooth rational surface has no nontrivial numerically trivial line bundles. The equality $\mu_* \mathcal{O}_{\widetilde{Y}} = \mathcal{O}_Y$ and the projection formula then give $K_Y \sim 0$. □

4 Albanese fibrations and effective nonvanishing

We devote this section to the proof of the following proposition.

Proposition 4.1. *Let X be a regular, geometrically non-normal, projective surface over k such that $H^0(X, \mathcal{O}_X) = k$, $K_X \equiv 0$ and $\dim \mathrm{Alb}_{X/k} > 0$. Then:*

1. *$p = 2$ or 3;*
2. *$6K_X \sim 0$ if $p = 3$, and $12K_X \sim 0$ if $p = 2$;*
3. *$\dim \mathrm{Alb}_{X/k} = 1$, and the Albanese morphism $\mathrm{alb}_{X/k}\colon X \to \mathrm{Alb}_{X/k}$ is a fibration; and*
4. *after a finite separable field extension k'/k and a choice of origin 0 on $E := (\mathrm{Alb}_{X/k})_{k'}$, there exists an integer $n \geq 1$ such that*

$$X_{k'} \simeq (E \times_{k'} C)/E[n], \qquad C := (\mathrm{alb}_{X/k})_{k'}^{-1}(0),$$

where $E[n]$ acts on E by translations and on $E \times_{k'} C$ diagonally.

We begin the proof with the following reductions and notation. First, we may assume, without loss of generality, that k is F-finite and separably closed. Set $k_1 := k^{1/p}$ and $X_1 := (X \times_k k_1)^{\nu}_{\mathrm{red}}$, and let $\phi\colon X_1 \to X$ be the natural morphism. Since X is geometrically non-normal, it is not para-abelian, so $\dim \mathrm{alb}_{X/k}(X) \neq 2$ by Proposition 2.24, and hence $\dim \mathrm{alb}_{X/k}(X) = 1$. By [Tan20, Theorem 1.1], we have $K_X \sim_{\mathbb{Q}} 0$. By Theorem 2.10, there is an effective divisor $\mathfrak{C} > 0$ such that

$$\phi^* K_X = K_{X_1} + (p-1)\mathfrak{C}. \tag{4.1}$$

Thus K_{X_1} is not pseudo-effective. Therefore, as in Construction 2.20, the MMP on the minimal resolution $\widetilde{X}_1$ of X terminates to a Mori fiber space $g\colon Z \to B$ and we have the following diagram

$$\begin{array}{ccccccc} & & & & \widetilde{X}_1 & & \\ & & & \swarrow{\scriptstyle \mu} & & \searrow{\scriptstyle \epsilon} & \\ X & \xleftarrow{\phi} & X_1 & & & & Z \\ & & & & & & \downarrow{\scriptstyle g} \\ & & & & & & B_1\,. \end{array} \tag{4.2}$$

Note that $H^0(X_1, \mathcal{O}_{X_1}) = H^0(\widetilde{X}_1, \mathcal{O}_{\widetilde{X}_1}) = H^0(Z, \mathcal{O}_Z) = H^0(B_1, \mathcal{O}_{B_1}) = k_1$. By Lemma 2.19 and Proposition 2.14, the exceptional curves of μ and ϵ, as well as the fibers of g, have trivial Albanese image. Hence the Albanese morphism of X_1 factors as

$$X_1 \xrightarrow{f_1} B_1 \longrightarrow \mathrm{Alb}_{X_1/k_1}\,.$$

Since the Albanese morphism is nonconstant, it follows that $p_a(B_1) > 0$. Let $X \to B_0 \to \mathrm{Alb}_{X/k}$ be the Stein factorization of $\mathrm{alb}_{X/k}$. Then we have the following commutative diagram

$$\begin{array}{ccccc} X_1 & \xrightarrow{f_1} & B_1 & \longrightarrow & \mathrm{Alb}_{X_1/k_1} \\ {\scriptstyle\phi}\downarrow & & \downarrow & & \downarrow \\ X & \longrightarrow & B_0 & \longrightarrow & \mathrm{Alb}_{X/k}\ . \end{array}$$

Let F be the generic fiber of f_1, which is a regular projective curve over $k_F := K(B_1)$. Since F is isomorphic to the generic fiber of g, it is an integral conic over k_F by Proposition 2.14. Restricting (4.1) to F and applying the adjunction formula, we obtain

$$0 = \deg_{k_F} K_F + (p-1)\mathfrak{C} \cdot_{k_F} F = -2 + (p-1)\mathfrak{C} \cdot_{k_F} F. \tag{4.3}$$

Thus $p = 2$ or 3, which proves Proposition 4.1 (1). Moreover, one of the following holds:

(a) $\mathfrak{C}$ has a component C_1 satisfying $C_1 \cdot_{k_F} F = 1$;
(b) $p = 2$ and the horizontal part of $\mathfrak{C}$ is an irreducible curve C_1 satisfying $C_1 \cdot_{k_F} F = 2$.

In Case (a), the induced morphism $\sigma\colon C_1 \to B_1$ is birational and we can write

$$(p-1)\mathfrak{C} = C_1 + C_1' + \mathfrak{C}_v, \tag{4.4}$$

where $\mathfrak{C}_v$ is the vertical part and $C_1 + C_1'$ is the horizontal part (possibly with $C_1' = C_1$). In Case (b), the induced morphism $\sigma\colon C_1 \to B_1$ is of degree two and we write $\mathfrak{C} = C_1 + \mathfrak{C}_v$.

Denote by C_0 the image of C_1 in X. Since ϕ is a homeomorphism, $\phi^* C_0 = \alpha C_1$ for some integer $\alpha \geq 1$.

Lemma 4.2. *Under the notation above:*

(i) *C_1 is smooth over $k_{C_1} := H^0(C_1, \mathcal{O}_{C_1})$ with $p_a(C_1) = 1$, and $C_1 \cdot \mathfrak{C}_v = 0$. In Case (a), σ is an isomorphism and $k_{C_1} = k_1$.*
(ii) *$C_1^2 = 0$ and C_1 is semi-ample.*

Proof. First, we claim that $C_1^2 \geq 0$. Otherwise $C_0^2 < 0$ (Corollary 2.6), and restricting $K_X + C_0 \equiv C_0$ to C_0^ν and applying the adjunction formula would then give

$$\deg K_{C_0^\nu} + \deg \mathrm{Diff}_{C_0^\nu} = C_0^2 < 0,$$

hence $p_a(C_0^\nu) = 0$, which is impossible because C_0^ν dominates the curve B_0 of positive geometric genus. Thus $C_1^2 \geq 0$.

Restricting (4.1) to C_1^ν and applying the adjunction formula, we obtain

$$K_{C_1^\nu} + C_1'|_{C_1^\nu} + \mathfrak{C}_v|_{C_1^\nu} + \mathrm{Diff}_{C_1^\nu} \sim_{\mathbb{Q}} 0 \text{ in Case (a)},$$

and

$$K_{C_1^\nu} + \mathfrak{C}_v|_{C_1^\nu} + \mathrm{Diff}_{C_1^\nu} \sim_{\mathbb{Q}} 0 \text{ in Case (b)}.$$

Since C_1 dominates B_1, it follows that $\deg K_{C_1^\nu} \geq 0$. All the other terms in the above equalities have non-negative degrees. In either case, we conclude that

$$\deg K_{C_1^\nu} = 0, \qquad \mathrm{Diff}_{C_1^\nu} = 0, \qquad C_1 \cdot \mathfrak{C}_v = 0,$$

and $C_1' \cdot C_1 = 0$ in case (a).

By Proposition 2.7, $\mathrm{Diff}_{C_1^\nu} = 0$ implies that C_1 is regular and that X_1 is regular along C_1. Since the composite $C_1 \to B_1 \to \mathrm{Alb}_{X_1/k_1}$ is nonconstant and $\deg K_{C_1^\nu} = 0$, it follows that C_1 is smooth over k_{C_1}. In Case (a), since C_1 and B_1 are normal, the finite birational morphism $\sigma\colon C_1 \to B_1$ is an isomorphism, so $k_{C_1} \simeq H^0(B_1, \mathcal{O}_{B_1}) = k_1$, which completes the proof of (i).

Next, we prove (ii). We first deal with Case (a). If $C_1' = C_1$, then $C_1^2 = C_1' \cdot C_1 = 0$. Thus, the image C_0 of C_1 on X also satisfies $C_0^2 = 0$. Since $K_X \sim_{\mathbb{Q}} 0$, by adjunction formula, we have

$$0 \sim_{\mathbb{Q}} (K_X + C_0)|_{C_0^\nu} \sim_{\mathbb{Q}} K_{C_0^\nu} + \mathrm{Diff}_{C_0^\nu}\ .$$

Since $\deg K_{C_0^\nu} \geq 0$, this implies that $\mathrm{Diff}_{C_0^\nu} = 0$, C_0 is regular, and $(K_X + C_0)|_{C_0} \sim K_{C_0} \sim 0$. Since $K_X \sim_{\mathbb{Q}} 0$, we have $C_0|_{C_0} \sim_{\mathbb{Q}} 0$ and $K_X \cdot C_0 = 0$. It follows from Proposition 2.8 that C_0 is semi-ample. Since $\phi^* C_0 = \alpha C_1$, the curve C_1 is semi-ample as well.

If $C_1' \neq C_1$, we have $C_1 \cdot C_1' = 0$. We have shown that $C_1^2 \geq 0$, and the same argument gives $C_1'^2 \geq 0$. By the Hodge index theorem, $C_1^2 = C_1'^2 = 0$. The argument above then shows that both C_1 and C_1' are semi-ample.

We next deal with Case (b). As in Case (a), it suffices to show that $C_1^2 = 0$. We take the base change

$$\begin{array}{ccc} X_2 := X_1 \times_{B_1} C_1 & \xrightarrow{\phi_2} & X_1 \\ {\scriptstyle f_2}\downarrow & & \downarrow{\scriptstyle f_1} \\ B_2 := C_1 & \xrightarrow[\sigma]{} & B_1. \end{array} \tag{4.5}$$

The fiber product X_2 is integral since $K(B_1)$ is algebraically closed in $K(X_1)$ and σ is either separable of purely inseparable of degree $p = 2$ (see [Tan18a, Lemma 3.3]). If σ is separable, the pullback of C_1 is $C_2 + C_2'$, and the induced morphism $C_2 \to B_2$ is birational.

Suppose first that σ is separable. Then $k_{C_1} = k_1$ since k_1 is separably closed. The Riemann–Hurwitz formula (Lemma 2.4) gives $K_{C_1} \sim \sigma^* K_{B_1} + R$ with $R \geq 0$. As $\deg K_{C_1} = 0$ and $\deg K_{B_1} \geq 0$, we get $R = 0$ and $p_a(B_1) = 1$. Hence σ is an étale double cover and $K_{B_1} \sim 0$. It follows that ϕ_2 is étale, so X_2 is normal, and that

$$K_{X_2} + C_2 + C_2' + \phi_2^* \mathfrak{C}_v \sim \phi_2^*(K_{X_1} + C_1 + \mathfrak{C}_v) \sim_{\mathbb{Q}} 0.$$

Restricting to C_2 and applying the adjunction formula, we obtain

$$K_{C_2} + C_2'|_{C_2} + \phi_2^* \mathfrak{C}_v|_{C_2} \sim_{\mathbb{Q}} 0.$$

Since $C_2 \simeq B_2 = C_1$ has arithmetic genus one, $\deg K_{C_2} = 0$. As $\phi_2^* \mathfrak{C}_v$ is f_2-vertical, we have $\phi_2^* \mathfrak{C}_v|_{C_2} \geq 0$. Therefore, we have $C_2 \cdot C_2' = 0$. Let ι be the nontrivial automorphism of C_1 over B_1. Then $\mathrm{id}_{X_1} \times \iota$ is an automorphism of X_2 exchanging C_2 and C_2', so $C_2^2 = C_2'^2$. Since $2C_1^2 = (\phi_2^* C_1)^2 = (C_2 + C_2')^2 = 2C_2^2$, we get $C_2^2 = C_1^2 \geq 0$, and the Hodge index theorem gives $C_2^2 = 0$. Hence $C_1^2 = 0$.

Suppose now that σ is inseparable. Let $\nu \colon X_2^\nu \to X_2$ be the normalization. The surface X_2 is Gorenstein in codimension one, where we have

$$K_{X_2} \sim \phi_2^* K_{X_1} + f_2^*(K_{C_1} - \sigma^* K_{B_1}).$$

The conductor formula, applied in codimension one, gives $K_{X_2^\nu} + \mathfrak{D} \sim \nu^* K_{X_2}$, where $\mathfrak{D} \geq 0$ is the conductor divisor of ν. Write $\nu^* \phi_2^* C_1 = mC_2$ and $\mathfrak{D} = dC_2 + \mathfrak{D}'$ with $C_2 \not\subset \operatorname{Supp} \mathfrak{D}'$. As $K_{C_1} \sim 0$ and $K_{X_1} \sim_{\mathbb{Q}} -C_1 - \mathfrak{C}_v$, we get

$$(K_{X_2^\nu} + C_2) + (m + d - 1)C_2 + \mathfrak{D}' + \nu^* \phi_2^* \mathfrak{C}_v + \nu^* f_2^* \sigma^* K_{B_1} \sim_{\mathbb{Q}} 0. \tag{4.6}$$

The coefficient $m + d - 1$ is positive. Indeed, $m \geq 1$, and if $d = 0$, then X_2 is normal at the generic point of $\nu(C_2)$. The non-reduced scheme-theoretic preimage $X_2 \times_{X_1} C_1 = C_1 \times_{B_1} C_1$ then forces $m \geq 2$. Restricting (4.6) to C_2^ν and applying the adjunction formula, we obtain

$$K_{C_2^\nu} + \mathrm{Diff}_{C_2^\nu} + (m + d - 1)C_2|_{C_2^\nu} + (\mathfrak{D}' + \nu^* \phi_2^* \mathfrak{C}_v + \nu^* f_2^* \sigma^* K_{B_1})|_{C_2^\nu} \sim_{\mathbb{Q}} 0.$$

Every term has nonnegative degree, since C_2^ν has nontrivial Albanese morphism, $m^2 C_2^2 = 2C_1^2 \geq 0$, $\nu^* \phi_2^* \mathfrak{C}_v$ is vertical over B_2, and $p_a(B_1) > 0$. Hence $C_2^2 = 0$, so $C_1^2 = 0$. This completes the proof. □

Proposition 4.3. *With the notation above, the surface X_1 is regular and $\mathfrak{C}_v = 0$.*

Proof. Let $h_1 \colon X_1 \to D_1$ be the fibration induced by the semi-ample divisor C_1 from Lemma 4.2 (ii). Let G be an arbitrary fiber of h_1. Since $G \equiv rC_1$ for some $r > 0$ and $C_1 \cdot F > 0$, we may write $G = nT + G'$, where T is an f_1-horizontal prime divisor and G' is an effective divisor such that $T \not\subset \operatorname{Supp} G'$. We also write $(p - 1)\mathfrak{C} = bT + \mathfrak{D}$, with $T \not\subset \operatorname{Supp} \mathfrak{D}$ and $b \geq 0$.

Since $G|_{T^\nu} \sim 0$ and $\phi^* K_X \sim_{\mathbb{Q}} 0$, we have

$$0 \sim_{\mathbb{Q}} (\phi^* K_X + G)|_{T^\nu} = (K_{X_1} + (p - 1)\mathfrak{C} + G)|_{T^\nu} = (K_{X_1} + T + (n + b - 1)T + \mathfrak{D} + G')|_{T^\nu}.$$

Applying the adjunction formula, we obtain

$$0 \sim_{\mathbb{Q}} K_{T^\nu} + \mathrm{Diff}_{T^\nu} + (n + b - 1)T|_{T^\nu} + (\mathfrak{D} + G')|_{T^\nu}. \tag{4.7}$$

The argument proving $C_1^2 \geq 0$ in Lemma 4.2 also gives $T^2 \geq 0$. Therefore, all the terms in (4.7) have nonnegative degrees. We conclude that $\mathrm{Diff}_{T^\nu} = 0$ and $G' \cdot T = 0$. Since the fiber G is connected, it follows from $G' \cdot T = 0$ that $G' = 0$. Hence, every fiber of h_1 has irreducible support. By Proposition 2.7,

$\mathrm{Diff}_{T^\nu} = 0$ implies that X_1 is regular along T. Since G was an arbitrary fiber and the fibers of h_1 cover X_1, the surface X_1 is regular.

Finally, it remains to show that $\mathfrak{C}_v = 0$. Suppose, to the contrary, that $\mathfrak{C}_v \neq 0$. Since C_1 is semi-ample and its induced fibration h_1 has irreducible fibers, $C_1 \cdot \mathfrak{C}_v = 0$ implies that $\mathfrak{C}_v$ is numerically equivalent to mC_1 for some $m > 0$. Then $\mathfrak{C}_v \cdot F = mC_1 \cdot F > 0$, contradicting $\mathfrak{C}_v \cdot F = 0$.

□

Lemma 4.4. *In the $p = 3$ case, we have*

$$2C_1|_{C_1} \sim 0.$$

Proof. As $\mathfrak{C}_v = 0$ by Proposition 4.3, we have

$$\phi^* K_X = K_{X_1} + 2C_1.$$

By Lemma 4.2, the curve $C_1 \simeq B_1$ is smooth over k_1 of arithmetic genus one. As X_1 is regular (Proposition 4.3), C_1 is an effective Cartier divisor, and Lemma 2.3 shows that X_1 is smooth over k_1 along C_1. Moreover, $H^0(X_1, \mathcal{O}_{X_1}) = k_1$ and $C_1^2 = 0$.

Put $N := \mathcal{O}_{C_1}(C_1)$, so $N \in \mathrm{Pic}^0(C_1)$. It suffices to show that $N^{\otimes 2} \simeq \mathcal{O}_{C_1}$. If C_1 is movable, there is an effective divisor $D \sim C_1$ that does not contain C_1. The equality $C_1 \cdot D = C_1^2 = 0$ shows that C_1 and D are disjoint, and hence $N \simeq \mathcal{O}_{C_1}(D) \simeq \mathcal{O}_{C_1}$. We may therefore assume that C_1 is not movable. The decomposition $C_1 = \mathfrak{F} + \mathfrak{M}$ in Theorem 2.10 then gives $\mathfrak{M} = 0$ and $\mathfrak{F} = C_1$, so $X \times_k k_1$ is reduced. Put $r := \dim_{k_1} \mathrm{Der}_k(k_1)$. Let $\mathscr{G} := T_{X_1/X}$ be the foliation associated with the height-one purely inseparable morphism ϕ, let $\mathscr{G}'$ be its saturation in $\mathrm{Der}_k(k_1) \otimes_{k_1} \mathcal{O}_{X_1}$, and denote the inclusion by $\iota \colon \mathscr{G} \hookrightarrow \mathscr{G}'$. Since $X \times_k k_1$ is reduced, Subsection 2.7 (see also [JW21, Remark 3.3]) gives

$$\mathscr{G}' = \mathrm{Der}_k(k_1) \otimes_{k_1} \mathcal{O}_{X_1} \simeq \mathcal{O}_{X_1}^{\oplus r},$$

and the zero divisor of $\det(\iota)$ is $\mathfrak{F} = C_1$.

Choose an open neighborhood $U \subset X_1$ of C_1 over which X_1 is smooth over k_1. Then the cotangent sequence for $U \to \operatorname{Spec} k_1 \to \operatorname{Spec} k$ is exact:

$$0 \longrightarrow \Omega_{k_1/k} \otimes_{k_1} \mathcal{O}_U \longrightarrow \Omega_{U/k} \longrightarrow \Omega_{U/k_1} \longrightarrow 0.$$

Since Ω_{U/k_1} is locally free, this sequence splits locally. Dualizing and using $\mathrm{Hom}_{k_1}(\Omega_{k_1/k}, k_1) \cong \mathrm{Der}_k(k_1)$ and $\mathscr{G}'|_U = \mathrm{Der}_k(k_1) \otimes_{k_1} \mathcal{O}_U$, we obtain the locally split exact sequence

$$0 \longrightarrow T_{X_1/k_1}|_U \longrightarrow T_{X_1/k}|_U \xrightarrow{\mathrm{res}} \mathscr{G}'|_U \longrightarrow 0. \tag{4.8}$$

The map res sends a local k-derivation D to its restriction $D|_{k_1}$. The composite

$$\mathscr{G}|_U \hookrightarrow T_{X_1/k}|_U \xrightarrow{\mathrm{res}} \mathscr{G}'|_U$$

equals $\iota|_U$. Dualizing the cotangent sequence for $X_1 \xrightarrow{\phi} X \to \operatorname{Spec} k$, we obtain an exact sequence

$$0 \longrightarrow \mathscr{G} \longrightarrow T_{X_1/k} \longrightarrow \mathcal{H}om_{\mathcal{O}_{X_1}}\big(\phi^* \Omega_{X/k}, \mathcal{O}_{X_1}\big).$$

Since X_1 is integral, the last sheaf is torsion-free. The kernel of a morphism into a torsion-free sheaf is saturated, since the quotient by the kernel embeds into the target. Hence, $\mathscr{G}$ is saturated in $T_{X_1/k}$.

The foliation $\mathscr{G}$ is reflexive and hence locally free on the regular surface X_1. Fix $x \in C_1$, let $\mathfrak{m} \subset \mathcal{O}_{X_1,x}$ be the maximal ideal, and let t be a local equation of C_1 at x. Since C_1 is regular, $t \notin \mathfrak{m}^2$. Let M be an $r \times r$ matrix over $\mathcal{O}_{X_1,x}$ representing ι_x. Since the zero divisor of $\det(\iota)$ is C_1, $\det M$ is a unit times t. If $r = 1$, then, after rescaling, $M = (t)$. If $r > 1$, then not all entries of M lie in $\mathfrak{m}$ (otherwise, $\det M \in \mathfrak{m}^r \subseteq \mathfrak{m}^2$, a contradiction). Using elementary row and column operations to clear a unit entry and inducting on r, we reduce M to $\mathrm{diag}(1, \ldots, 1, t)$. In either case, the cokernel of ι is a line bundle A on C_1, which induces an exact sequence near C_1:

$$0 \longrightarrow \mathscr{G} \xrightarrow{\iota} \mathscr{G}' \longrightarrow i_* A \longrightarrow 0, \qquad i \colon C_1 \hookrightarrow X_1. \tag{4.9}$$

Consider the exact sequence

$$0 \longrightarrow \mathcal{O}_{X_1}(-C_1) \longrightarrow \mathcal{O}_{X_1} \longrightarrow i_* \mathcal{O}_{C_1} \longrightarrow 0.$$

Locally, the first map is the multiplication by a local equation of C_1, which annihilates $i_* A$. Tensoring this sequence with $i_* A$, we obtain an isomorphism of sheaves on C_1:

$$\mathrm{Tor}_1^{\mathcal{O}_{X_1}}(\mathcal{O}_{C_1}, i_* A) \simeq A \otimes \mathcal{O}_{C_1}(-C_1) = A \otimes N^{-1}.$$

Tensoring (4.9) with $\mathcal{O}_{C_1}$ and using the fact that $\mathscr{G}'$ is locally free, we obtain

$$0 \longrightarrow \operatorname{Tor}_1^{\mathcal{O}_{X_1}}(i_*A, \mathcal{O}_{C_1}) \cong A \otimes N^{-1} \longrightarrow \mathscr{G}|_{C_1} \longrightarrow \mathscr{G}'|_{C_1} \simeq \mathcal{O}_{C_1}^{\oplus r} \longrightarrow A \longrightarrow 0. \tag{4.10}$$

In particular, A is globally generated. Put $P := A \otimes N^{-1}$. Since $\mathscr{G}$ is saturated in $T_{X_1/k}$, the restriction $\mathscr{G}|_{C_1} \to T_{X_1/k}|_{C_1}$ is injective. The exact sequence (4.10) and the restriction of (4.8) to C_1 fit into the following commutative diagram with exact rows:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & P & \longrightarrow & \mathscr{G}|_{C_1} & \xrightarrow{\iota|_{C_1}} & \mathscr{G}'|_{C_1} & \longrightarrow & A & \longrightarrow & 0 \\
 & & {\scriptstyle\theta}\downarrow & & \downarrow & & \| & & \downarrow & & \\
0 & \longrightarrow & T_{X_1/k_1}|_{C_1} & \longrightarrow & T_{X_1/k}|_{C_1} & \xrightarrow{\text{res}} & \mathscr{G}'|_{C_1} & \longrightarrow & 0 & &
\end{array}$$

This yields an injection $\theta\colon P \to T_{X_1/k_1}|_{C_1}$.

Since C_1 is a smooth curve of genus one, we have $T_{C_1/k_1} \simeq \mathcal{O}_{C_1}$, and the tangent–normal exact sequence is

$$0 \longrightarrow T_{C_1/k_1} \simeq \mathcal{O}_{C_1} \longrightarrow T_{X_1/k_1}|_{C_1} \xrightarrow{\rho} N = \mathcal{O}_{C_1}(C_1) \longrightarrow 0.$$

According as $\rho \circ \theta$ is nonzero or zero, θ induces a nonzero map from P to N or to $\mathcal{O}_{C_1}$. Since both targets have degree zero, $\deg P \leq 0$. On the other hand, $\deg P = \deg A \geq 0$, since A is globally generated. Hence, we have $A \simeq \mathcal{O}_{C_1}$ and $P \simeq N^{-1}$. The same nonzero map between degree-zero line bundles is now an isomorphism. Therefore, we have either $N^{-1} \cong N$ or $N \cong \mathcal{O}_{C_1}$. Thus $N^{\otimes 2} \simeq \mathcal{O}_{C_1}$ in both cases. □

Remark 4.5. In the non-movable case of the proof of Lemma 4.4, we have the following correspondence:

ϕ^*C_0	C_1	Meaning	$\rho \circ \theta$	Conclusion
C_1	$\mathscr{G}$-invariant	$\mathscr{G}$ tangent to C_1	0	$N \simeq \mathcal{O}_{C_1}$
pC_1	Non-invariant	$\mathscr{G}$ generically transverse to C_1	Nonzero	$N^{\otimes 2} \simeq \mathcal{O}_{C_1}$

We now verify that C_1 is $\mathscr{G}$-invariant if and only if $\rho \circ \theta = 0$. Since $A \simeq \mathcal{O}_{C_1}$ and $H^0(C_1, \mathcal{O}_{C_1}) = k_1$, the quotient $\mathscr{G}'|_{C_1} \twoheadrightarrow A$ has constant coefficients. At the generic point of C_1, choose a local equation t, a basis $\delta_1, \ldots, \delta_r$ of $\operatorname{Der}_k(k_1)$ adapted to the quotient $\mathscr{G}'|_{C_1} \twoheadrightarrow A$, and generators $D_1, \ldots, D_r$ of $\mathscr{G}$ such that $\iota(D_i) = \delta_i$ for $i < r$ and $\iota(D_r) = t\delta_r$. Lie-closedness gives $[D_i, D_r] \in \mathscr{G}$, and

$$\iota([D_i, D_r]) = D_i(t)\delta_r + t[\delta_i, \delta_r].$$

Since $[D_i, D_r] \in \mathscr{G}$, reducing modulo t and projecting to A gives zero on the left and the image of $D_i(t)\delta_r$ on the right. As the image of δ_r generates A, we obtain $D_i(t) \in (t)$ for $i < r$. Hence C_1 is invariant precisely when $D_r(t) \in (t)$, or equivalently $\rho \circ \theta = 0$, since $D_r \bmod t$ generates P.

Proposition 4.6. *With the notation above, suppose that $p = 3$. Then $6K_X \sim 0$. Moreover, if C_1 is movable or $\phi^*C_0 = C_1$, then $3K_X \sim 0$.*

Proof. Since X_1 is regular, we have $\mu = \mathrm{id}$, so $f_1 = g \circ \epsilon$. We have the following commutative diagram

$$\begin{array}{ccc}
X & \xleftarrow{\phi} & X_1 \\
 & & \uparrow \quad \searrow^{f_1} \\
 & & C_1 \xrightarrow{\sigma} B_1\,.
\end{array}$$

The morphism σ is an isomorphism. Since C_1 is a regular curve of arithmetic genus one, adjunction gives $(K_{X_1} + C_1)|_{C_1} \sim K_{C_1} \sim 0$, so

$$\phi^*K_X|_{C_1} = (K_{X_1} + 2C_1)|_{C_1} \sim C_1|_{C_1}.$$

By Lemma 2.21 (2), there is a line bundle M on B_1 such that $\phi^*K_X \simeq f_1^*M$. Restricting to C_1 gives $\sigma^*M \simeq \mathcal{O}_{C_1}(C_1)$. Lemma 4.4 and the isomorphism σ imply that $2M \sim 0$, hence $2\phi^*K_X \sim 0$. Lemma 2.1 (2) gives $6K_X \sim 0$.

Under either additional condition, $C_1|_{C_1} \sim 0$ by the proof of Lemma 4.4 and Remark 4.5, so $M \sim 0$ and $3K_X \sim 0$. □

Proposition 4.7. *With the notation above, suppose that $p = 2$. Then $12K_X \sim 0$.*

Proof. Since X_1 is regular, we have $\mu = \mathrm{id}$ and $f_1 = g \circ \epsilon$. In Case (a), we have $\phi^* K_X = K_{X_1} + C_1 + C_1'$ and $\sigma\colon C_1 \to B_1$ is an isomorphism. If $C_1 \neq C_1'$, we obtain $C_1 \cap C_1' = \emptyset$ by $C_1 \cdot C_1' = 0$. Since C_1 is a regular curve of arithmetic genus one, it follows that

$$\phi^* K_X|_{C_1} = (K_{X_1} + C_1 + C_1')|_{C_1} = K_{C_1} \sim 0.$$

As in the proof of Proposition 4.6, Lemma 2.21 (2) gives $\phi^* K_X \simeq f_1^* M$ for a line bundle M on B_1, and $\sigma^* M \sim 0$ forces $M \sim 0$. Hence $\phi^* K_X \sim 0$, and Lemma 2.1(2) gives $2K_X \sim 0$.

Next we consider the case $C_1 = C_1'$. We have the following commutative diagram:

$$\begin{array}{ccc} X & \xleftarrow{\phi} & X_1 \\ \uparrow & & \uparrow \\ C_0 & \xleftarrow[\psi]{} & C_1 \end{array}$$

We have shown that C_1 and its image C_0 in X are regular curves of arithmetic genus one. We have $(K_{X_1} + C_1)|_{C_1} = K_{C_1} \sim 0$ and $(K_X + C_0)|_{C_0} = K_{C_0} \sim 0$. Moreover, we have $\phi^* C_0 = \alpha C_1$ with $\alpha \in \{1, 2\}$. It follows that

$$\phi^*(K_X + C_0)|_{C_1} = (K_{X_1} + (\alpha + 2)C_1)|_{C_1} = \psi^*((K_X + C_0)|_{C_0}) \sim 0.$$

Thus $(\alpha + 1)C_1|_{C_1} \sim 0$, and hence $6\phi^* K_X|_{C_1} \sim 0$. As above, writing $6\phi^* K_X \simeq f_1^* M$ by Lemma 2.21 (2), we get $\sigma^* M \sim 0$, hence $M \sim 0$ and $6\phi^* K_X \sim 0$. By Lemma 2.1(2), we obtain $12K_X \sim 0$.

In Case (b), we have $\phi^* K_X = K_{X_1} + C_1$ and $\sigma\colon C_1 \to B_1$ is a morphism of degree two. Since C_1 is a regular curve of genus one, we obtain $\phi^* K_X|_{C_1} = (K_{X_1} + C_1)|_{C_1} = K_{C_1} \sim 0$. Writing $\phi^* K_X \simeq f_1^* M$ by Lemma 2.21 (2), we get $\sigma^* M \sim 0$, so $2M \sim 0$ by Lemma 2.1 (1) since σ has degree two. Hence $2\phi^* K_X \sim 0$. Therefore, we obtain $4K_X \sim 0$ by Lemma 2.1(2).

We conclude that $12K_X \sim 0$ if $p = 2$. □

Lemma 4.8. *We have* $\dim \mathrm{Alb}_{X/k} = 1$. *Moreover, there is a fibration* $h\colon X \to D$ *onto a regular projective curve whose generic fiber* Γ *satisfies* $p_a(\Gamma) = 1$, *and* $\mathrm{alb}_{X/k}$ *does not factor through* h.

Proof. Set $A := \mathrm{Alb}_{X/k}$ and $\mathrm{alb} := \mathrm{alb}_{X/k}$. Since the Albanese morphism of X_1 factors through $f_1\colon X_1 \to B_1$, the universal property identifies Alb_{X_1/k_1} with Alb_{B_1/k_1}. The normalization of $(B_1 \otimes_{k_1} \bar{k})_{\mathrm{red}}$ is an elliptic curve, since it is dominated by the elliptic curve $(C_1 \otimes_{k_1} \bar{k})_{\mathrm{red}}$ and maps nonconstantly to an abelian variety. Hence $\dim \mathrm{Alb}_{B_1/k_1} = 1$ by Lemma 2.23 (3). By Lemma 2.23, we see that $\dim A = 1$.

By the proof of Lemma 4.2, the curve C_0 satisfies $C_0^2 = 0$ and is semi-ample. Let $h\colon X \to D$ be the fibration induced by C_0. Since $C_0 \neq 0$ and $C_0^2 = 0$, the base D is a curve, and it is regular because X is normal. The generic fiber Γ of h is a regular curve with $H^0(\Gamma, \mathcal{O}_\Gamma) = K(D)$, and $p_a(\Gamma) = 1$ because $K_X \equiv 0$ and the adjunction formula give $\deg K_\Gamma = 0$. Finally, alb does not factor through h, since the h-vertical curve C_0 has nonconstant image under alb. □

Proof of Proposition 4.1. Assertion (1) follows from (4.3), while assertion (2) follows from Propositions 4.6 and 4.7. Lemma 4.8 shows that $\dim \mathrm{Alb}_{X/k} = 1$ and provides a fibration $h\colon X \to D$. Since k is assumed to be separably closed, we may choose an origin on $\mathrm{Alb}_{X/k}$, so that it is an elliptic curve. Since $K_X \equiv 0$, the divisor $K_{X/D}$ is h-numerically trivial, hence h-nef. Assertions (3) and (4) now follow from Lemma 2.25. □

5 Torsion bounds with trivial Albanese variety

We study numerically trivial line bundles on regular, geometrically non-normal surfaces X with $-K_X$ nef and trivial Albanese variety. In this section, we prove $pL \sim 0$ for $p \geq 3$ and $4L \sim 0$ for $p = 2$.

5.1 Auxiliary results

For a numerically trivial line bundle L on a regular del Pezzo surface Z, Proposition 2.13 gives $3L \sim 0$ if $p = 3$ and $4L \sim 0$ if $p = 2$. The following two lemmas strengthen these bounds under additional hypotheses.

Lemma 5.1. *Let* Z *be a regular projective del Pezzo surface over a field* k *of characteristic* 3, *with* $H^0(Z, \mathcal{O}_Z) = k$. *Assume that one of the following holds:*

(a) Z *is geometrically integral*

(b) Z *admits a non-reduced effective Cartier divisor* D *such that* $D \equiv -K_Z$.

Then every numerically trivial line bundle on Z is trivial.

Proof. Under assumption (b), we first show that Z is geometrically normal. Let A be a prime component of D with multiplicity at least two, and put $R := D - 2A \geq 0$. Suppose the contrary that Z is geometrically non-normal. Set

$$W := (Z \times_k \overline{k})^{\nu}_{\mathrm{red}}, \qquad \psi \colon W \to Z,$$

where $\bar{k}$ is the algebraic closure of k. By [PW22, Theorem 4.1], there is an effective integral divisor C_W such that

$$K_W + 2C_W \sim \psi^* K_Z,$$

and (W, C_W) is either $(\mathbb{P}^2, L)$ or (S_d, F) for $d \geq 2$. Here L is a line in $\mathbb{P}^2$. In the second case, let $\sigma \colon \mathbb{F}_d \to S_d \simeq \mathbb{P}(1,1,d)$ contract the negative section E, and set $F := \sigma(\widetilde{F})$ for a fiber $\widetilde{F}$ of $\mathbb{F}_d \to \mathbb{P}^1$. Note that $\mathrm{Cl}(S_d) = \mathbb{Z}[F]$. Taking $d = 1$ in the first case, we may therefore write $W = \mathbb{P}(1,1,d)$ for $d \geq 1$ and let H denote the generator represented by L or F, respectively. Then

$$K_W \sim -(d+2)H, \qquad C_W \sim H, \qquad \mathrm{Pic}(W) = \mathbb{Z}[dH], \qquad \psi^*(-K_Z) \sim dH.$$

Since $\psi^* A$ is a nonzero effective Cartier divisor and $\psi^* R$ is an effective Cartier divisor, we have

$$\psi^* A \sim adH, \qquad \psi^* R \sim rdH$$

for integers $a \geq 1$ and $r \geq 0$. Since $-K_Z \equiv D = 2A + R$, comparison of numerical classes after pullback shows that $1 = 2a + r \geq 2$, a contradiction.

Let L be a numerically trivial line bundle on Z. In both cases of the hypothesis, Z is geometrically integral. Hence $H^1(Z, \mathcal{O}_Z) = 0$ by [BM24, Proposition 4.10]. Serre duality and ampleness of $-K_Z$ give

$$H^2(Z, L) = H^0(Z, \omega_Z \otimes L^{-1})^{\vee} = 0, \qquad H^2(Z, \mathcal{O}_Z) = 0.$$

Riemann–Roch yields $\chi(Z, L) = \chi(Z, \mathcal{O}_Z) = 1$, so $H^0(Z, L) \neq 0$. Thus $L \simeq \mathcal{O}_Z$. □

Lemma 5.2. *Let Z be a regular del Pezzo surface over a field k of characteristic 2, with $H^0(Z, \mathcal{O}_Z) = k$. If there exists an effective Cartier divisor $A > 0$ with $A \equiv -K_Z$, then every numerically trivial line bundle L on Z satisfies $2L \sim 0$.*

Proof. Let L be a numerically trivial line bundle on Z, and set

$$\psi \colon W := (Z \times_k \bar{k})^{\nu}_{\mathrm{red}} \longrightarrow Z.$$

Then W is a rational surface and thus $\psi^* L \sim 0$. By [BT22, Theorems 3.7], we have $\ell_F(Z/k) \leq 2$. If $\ell_F(Z/k) \leq 1$, then $2L \sim 0$ holds. We may therefore assume that $\ell_F(Z/k) = 2$. By [BM24, Theorem 3.5(4)] and [Tan24, Theorem 4.6(3)],

$$W \simeq \mathbb{P}(1,1,d), \qquad d \in \{1, 2, 4\}, \qquad \psi^*(-K_Z) \sim dH,$$

where H is the positive generator of the Weil divisor class group of W.

We distinguish the following two cases:

(a) There exists a Cartier divisor $M \equiv 2K_Z$ with $H^1(Z, M) \neq 0$.
(b) We have $H^1(Z, M) = 0$ for every Cartier divisor $M \equiv 2K_Z$.

Case (a). Fix a Cartier divisor M as in (a). By Serre vanishing,

$$H^1(Z, 2^n M) \simeq H^1(Z, K_Z - 2^n M)^{\vee} = 0 \quad \text{for } n \gg 0.$$

Since $H^1(Z, M) \neq 0$, we may choose $q = 2^e$ with $e \geq 0$ maximal such that $H^1(Z, qM) \neq 0$. Set $\mathscr{L} = \mathcal{O}_Z(qM)$. Following [Ek88, I, (1.5)], define $\alpha_{\mathscr{L}}$ by the exact sequence of fppf sheaves

$$0 \longrightarrow \alpha_{\mathscr{L}} \longrightarrow \mathscr{L} \xrightarrow{F} \mathscr{L}^{\otimes 2} \longrightarrow 0, \tag{5.1}$$

where F is the relative Frobenius morphism. The group scheme $\alpha_{\mathscr{L}}$ is finite flat of degree two and locally isomorphic to $\alpha_2 = \mathrm{Spec}(\mathcal{O}_Z[\epsilon]/(\epsilon^2))$. An $\alpha_{\mathscr{L}}$-torsor is a scheme with an $\alpha_{\mathscr{L}}$-action that is fppf locally isomorphic to the group acting on itself by translation. Such torsors are classified by $H^1_{\mathrm{fppf}}(Z, \alpha_{\mathscr{L}})$. The long exact cohomology sequence of (5.1) gives

$$H^0(Z, \mathscr{L}^{\otimes 2}) \longrightarrow H^1_{\mathrm{fppf}}(Z, \alpha_{\mathscr{L}}) \longrightarrow H^1(Z, \mathscr{L}) \xrightarrow{F} H^1(Z, \mathscr{L}^{\otimes 2}) = 0.$$

Thus a nonzero class in $H^1(Z,\mathscr{L})$ lifts to a nontrivial $\alpha_{\mathscr{L}}$-torsor $\pi\colon T\to Z$. Over an affine open $U=\operatorname{Spec} R$ trivializing $\mathscr{L}$, this torsor has the form $T|_U=\operatorname{Spec}(R[t]/(t^2-b))$ for some $b\in R$; see [Ek88, I, p. 106]. Hence π is finite flat and purely inseparable of degree two, and T is Gorenstein because Z is regular. As in [Ek88, I, Proposition 1.7], T is reduced and thus integral, and we have

$$K_T\sim\pi^*(K_Z+qM). \tag{5.2}$$

Put $\mathscr{L}_W:=\psi^*\mathscr{L}\simeq\mathcal{O}_W(-2qdH)$. Let $h\colon\mathbb{P}^2\to W$ be the finite morphism induced by the graded free extension

$$\bar{k}[x,y,z]\hookrightarrow\bar{k}[x,y,t],\qquad z\longmapsto t^d,\qquad \deg z=d,\quad \deg x=\deg y=\deg t=1.$$

Then $\mathcal{O}_W$ is a direct summand of $h_*\mathcal{O}_{\mathbb{P}^2}$ and $h^*\mathscr{L}_W\simeq\mathcal{O}_{\mathbb{P}^2}(-2qd)$. Since the inclusion $\mathcal{O}_W\hookrightarrow h_*\mathcal{O}_{\mathbb{P}^2}$ splits, we have

$$H^1(W,\mathscr{L}_W)\hookrightarrow H^1(W,\mathscr{L}_W\otimes h_*\mathcal{O}_{\mathbb{P}^2})\simeq H^1(W,h_*h^*\mathscr{L}_W)\simeq H^1(\mathbb{P}^2,h^*\mathscr{L}_W)=0.$$

Thus $H^1(W,\mathscr{L}_W)=0$. Since also $H^0(W,\mathscr{L}_W^{\otimes 2})=0$, the cohomology sequence of (5.1), after base change to W, gives $H^1_{\mathrm{fppf}}(W,\alpha_{\mathscr{L}_W})=0$, so the pulled-back torsor is trivial. Hence $(T\times_Z W)_{\mathrm{red}}\simeq W$, and its finite map to $(T\times_k\bar{k})_{\mathrm{red}}$ is birational. Since W is normal, it is the normalization of $(T\times_k\bar{k})_{\mathrm{red}}$. The resulting morphisms fit into the commutative diagram

$$\begin{array}{ccccc}
(T\times_k\bar{k})^{\nu}_{\mathrm{red}} & \xrightarrow{\sim} & (T\times_Z W)_{\mathrm{red}} & \xrightarrow{\sim} & W=(Z\times_k\bar{k})^{\nu}_{\mathrm{red}}\\
\downarrow & \swarrow & & & \downarrow{\scriptstyle\psi}\\
T & & \xrightarrow[\pi]{} & & Z\,.
\end{array}$$

Let $\nu\colon T^\nu\to T$ be the normalization and put $k_T=H^0(T^\nu,\mathcal{O}_{T^\nu})$. The morphism $T^\nu\to Z$ has height one, so T^ν is $\mathbb{Q}$-factorial. For $a\in k_T$, we have $a^2\in K(Z)$ algebraic over k, hence $a^2\in k$; thus k_T/k is purely inseparable of height at most one. Identifying W with the normalization of $(T^\nu\times_{k_T}\bar{k})_{\mathrm{red}}$, let $\theta\colon W\to T^\nu$ denote the natural morphism. The conductor formula [PW22, Lemma 2.14] and the canonical base-change formula [Tan21, Theorem 3.16] give

$$K_{T^\nu}+\Gamma\sim\nu^*K_T,\qquad K_W+D\sim_{\mathbb{Q}}\theta^*K_{T^\nu},\qquad \Gamma,D\geq 0,$$

where D is nonzero if T^ν/k_T is not geometrically normal. Since $K_W\sim-(d+2)H$, equation (5.2) gives

$$D+\theta^*\Gamma\sim_{\mathbb{Q}}(2-2qd)H. \tag{5.3}$$

Since D and Γ are effective, (5.3) shows that $2-2qd\geq 0$, so $q=d=1$. The effective divisor $D+\theta^*\Gamma$ is then numerically trivial, hence zero, so $D=\Gamma=0$. The conductor is zero in codimension one, and T is S_2 because it is Gorenstein; hence $T=T^\nu$ is normal. The condition $D=0$ implies that T is geometrically normal over k_T, so $T\times_{k_T}\bar{k}\simeq W\simeq\mathbb{P}^2$. Therefore π^*L is trivial for every numerically trivial L, and thus $2L\sim 0$.

Case (b). In this case $H^1(Z,M)=0$ for every Cartier divisor $M\equiv 2K_Z$. By assumption, there exists an effective Cartier divisor A with $A\equiv -K_Z$. By Serre duality, $H^1(Z,\mathcal{O}_Z(A))=H^1(Z,\mathcal{O}(K_Z-A))^\vee=0$ since $K_Z-A\equiv 2K_Z$, and $H^2(Z,\mathcal{O}_Z(A))=H^0(Z,\mathcal{O}(K_Z-A))^\vee=0$. Hence

$$\chi(\mathcal{O}_Z(A))=h^0(Z,\mathcal{O}_Z(A))>0.$$

For a numerically trivial line bundle L on Z, Riemann–Roch and Serre duality give

$$\chi(\omega_Z^{-1}\otimes L)=\chi(\mathcal{O}_Z(A))>0,\qquad H^2(Z,\omega_Z^{-1}\otimes L)=0.$$

Hence $h^0(Z,\omega_Z^{-1}\otimes L)\geq\chi(\omega_Z^{-1}\otimes L)>0$. Choose $A_L\in|\omega_Z^{-1}\otimes L|$. Each prime component of A_L is Cartier because Z is regular, and its pullback is linearly equivalent to a positive integral multiple of dH. Since $\psi^*A_L\sim dH$, the divisor A_L consists of a single prime component with multiplicity one, hence is nonzero and integral. Adjunction and Riemann–Roch give, respectively,

$$L|_{A_L}\simeq\omega_{A_L},\qquad \chi(\mathcal{O}_{A_L})=-\frac{1}{2}A_L\cdot(A_L+K_Z)=0. \tag{5.4}$$

By Serre duality and $\chi(\mathcal{O}_{A_L})=0$, we have

$$h^0(A_L,\omega_{A_L})=h^1(A_L,\mathcal{O}_{A_L})=h^0(A_L,\mathcal{O}_{A_L})\geq 1.$$

Thus ω_{A_L} has degree zero and a nonzero section, so it is trivial; in particular $L|_{A_L} \cong \mathcal{O}_{A_L}$. Let $2A_L \subset Z$ denote the closed subscheme defined by the ideal $\mathcal{O}_Z(-2A_L)$. The closed immersion $A_L \hookrightarrow 2A_L$ induces the restriction homomorphism $\mathrm{Pic}(2A_L) \to \mathrm{Pic}(A_L)$, sending the class of a line bundle $\mathcal{N}$ to that of $\mathcal{N}|_{A_L}$. The ideal sheaf of A_L in $2A_L$ is

$$\mathcal{I} = \mathcal{O}_Z(-A_L)/\mathcal{O}_Z(-2A_L) \subset \mathcal{O}_{2A_L}.$$

Locally, if A_L is defined by $f \in R$, then $\mathcal{I}$ corresponds to the ideal $(f)/(f^2)$ in $R/(f^2)$, whose square is zero. Let $\mathcal{N}$ be a line bundle on $2A_L$ whose restriction to A_L is trivial. Fix a trivialization $\mathcal{N}|_{A_L} \simeq \mathcal{O}_{A_L}$. Choose local trivializations of $\mathcal{N}$ extending this trivialization, so that its transition functions have the form $1 + u_{ij}$ with u_{ij} local sections of $\mathcal{I}$. Since $\mathcal{I}^2 = 0$ and the characteristic is two, these transition functions satisfy

$$(1 + u_{ij})^2 = 1 + 2u_{ij} + u_{ij}^2 = 1.$$

Thus $\mathcal{N}^{\otimes 2} \simeq \mathcal{O}_{2A_L}$. Taking $\mathcal{N} = L|_{2A_L}$, whose restriction to A_L is trivial as obtained before, we obtain $L^{\otimes 2}|_{2A_L} \simeq \mathcal{O}_{2A_L}$. Finally, the exact sequence

$$0 \longrightarrow \mathcal{O}_Z(2K_Z) \longrightarrow L^{\otimes 2} \longrightarrow \mathcal{O}_{2A_L} \longrightarrow 0 \tag{5.5}$$

and $H^1(Z, \mathcal{O}_Z(2K_Z)) = 0$ give a nonzero section of $L^{\otimes 2}$ and hence $2L \sim 0$. □

Lemma 5.3. *Let k be a separably closed field of characteristic 3, and let C be a regular, geometrically integral projective curve over k with $H^0(C, \mathcal{O}_C) = k$ and $p_a(C) = 1$. If C is not smooth over k, then*

$$C \simeq \operatorname{Proj} k[x, y, z]/(y^2z - x^3 - az^3) \qquad \text{for some } a \in k \setminus k^3.$$

Proof. Since C is regular with $p_a(C) = 1$, we have $\deg \omega_C = 0$ and $h^0(C, \omega_C) = h^1(C, \mathcal{O}_C) = 1$; a nonzero section of a degree-zero line bundle on an integral projective curve vanishes nowhere, so $\omega_C \simeq \mathcal{O}_C$.

The smooth locus of C is a nonempty open subscheme, and it has a k-rational point P because k is separably closed. By [Tan21, Corollary 11.12(1)], the linear system $|3P|$ embeds C into $\mathbb{P}^2_k$ as a cubic. In Weierstrass coordinates, its equation has the form

$$y^2z + b_1xyz + b_3yz^2 = x^3 + b_2x^2z + b_4xz^2 + b_6z^3, \qquad b_i \in k.$$

Since 2 is invertible in k, set $Y = y + \frac{1}{2}(b_1x + b_3z)$. The identity

$$y^2z + (b_1x + b_3z)yz = Y^2z - \frac{1}{4}(b_1x + b_3z)^2z$$

eliminates the terms linear in Y. Renaming Y as y and the resulting coefficients as a_2, a_4, a_6, we obtain

$$C = \{y^2z = x^3 + a_2x^2z + a_4xz^2 + a_6z^3\}.$$

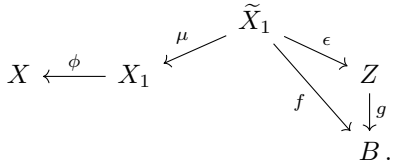
The point $(0 : 1 : 0)$ is smooth, and a point of $C \cap \{z \neq 0\}$ is singular over $\overline{k}$ if and only if $y = 0$ and x is a multiple root of $f(x) := x^3 + a_2x^2 + a_4x + a_6$. Such a root exists because C is not smooth. If it were not a triple root, it would be the root of the greatest common divisor of f and f' in $k[x]$, hence would lie in k, and C would have a singular k-rational point, contradicting regularity. Therefore $f = (x - x_0)^3 = x^3 - x_0^3$, so $a_2 = a_4 = 0$ and $a := a_6$; moreover $a \notin k^3$, since otherwise the singular point would be k-rational. □

5.2 The case $K_X \equiv 0$

Proposition 5.4. *Let X be a regular, geometrically non-normal projective surface over a field k of characteristic $p > 0$ such that $H^0(X, \mathcal{O}_X) = k$, $K_X \equiv 0$ and $\dim \mathrm{Alb}_{X/k} = 0$. Then every numerically trivial line bundle L on X satisfies*

$$pL \sim 0 \quad \text{if } p \geq 3, \qquad 4L \sim 0 \quad \text{if } p = 2.$$

Proof. By descending (X, L) to a model satisfying the same hypotheses over a finitely generated subfield and then passing to its separable closure, we may assume that k is F-finite and separably closed. Set $k_1 = k^{1/p}$ and $X_1 = (X \times_k k_1)^{\nu}_{\mathrm{red}}$. Construction 2.20 gives



Here μ is the minimal resolution, ϵ is a $K_{\widetilde{X}_1}$-MMP contraction, g is a Mori fiber space, and $f = g \circ \epsilon$. Note that

$$H^0(X_1, \mathcal{O}_{X_1}) = H^0(\widetilde{X}_1, \mathcal{O}_{\widetilde{X}_1}) = H^0(Z, \mathcal{O}_Z) = H^0(B, \mathcal{O}_B) = k_1.$$

In this proof, intersection numbers are computed over k on X and over k_1 on X_1, $\widetilde{X}_1$, Z, and their further birational models. Theorem 2.10 and (2.7) give

$$K_{X_1} + (p-1)\mathfrak{C} \sim \phi^* K_X, \quad \Delta := (p-1)\widetilde{\mathfrak{C}} + \sum_i m_i E_i, \quad K_{\widetilde{X}_1} + \Delta \sim \mu^*\phi^* K_X \equiv 0, \tag{5.6}$$

where $\mathfrak{C} > 0$ is an effective integral divisor, $\widetilde{\mathfrak{C}}$ is its strict transform, the E_i are μ-exceptional, and $m_i \in \mathbb{Z}_{\geq 0}$. Lemma 2.1(2) reduces the assertion to

$$\phi^* L \sim 0 \quad \text{if } p \geq 3, \qquad 2\phi^* L \sim 0 \quad \text{if } p = 2. \tag{5.7}$$

The del Pezzo case. Suppose that $\dim B = 0$. By Lemma 2.21(1), there is a numerically trivial line bundle L_Z on Z such that

$$\mu^*\phi^* L \simeq \epsilon^* L_Z. \tag{5.8}$$

By injectivity of $\mu^*\colon \operatorname{Pic}(X_1) \to \operatorname{Pic}(\widetilde{X}_1)$, to prove (5.7) it suffices to show that $L_Z \sim 0$ for $p \geq 3$ and $2L_Z \sim 0$ for $p = 2$.

For $p \geq 5$, Proposition 2.13 gives $L_Z \sim 0$.

Suppose that $p = 3$. If Z is geometrically integral, then $L_Z \sim 0$ by Lemma 5.1. If Z is not geometrically integral, then Z, and hence X, is geometrically non-reduced, so $\mathfrak{C} = \mathfrak{M} + \mathfrak{F}$ has a nonzero movable part. In this case, set

$$A = \epsilon_* \widetilde{\mathfrak{M}} > 0, \qquad R = \epsilon_* \left(2\widetilde{\mathfrak{F}} + \sum_i m_i E_i \right) \geq 0.$$

Pushing forward (5.6) to Z gives $-K_Z \equiv 2A + R$. So in this case, we also have $L_Z \sim 0$ by Lemma 5.1.

Suppose now that $p = 2$. Set $A := \epsilon_* \Delta$. Since Z is regular, A is an effective Cartier divisor, and pushing forward (5.6) gives $A \equiv -K_Z$. The ampleness of $-K_Z$ implies $A > 0$. Lemma 5.2 therefore gives $L_Z^{\otimes 2} \simeq \mathcal{O}_Z$.

The ruled case. Suppose that $\dim B = 1$. Since $\operatorname{Alb}_{X/k}$ is trivial, B has geometric genus zero. By Lemma 2.21(2), there is a degree-zero line bundle L_B on B such that

$$\mu^*\phi^* L \simeq f^* L_B. \tag{5.9}$$

We must prove $L_B \sim 0$ for $p \geq 3$ and $2L_B \sim 0$ for $p = 2$. Lemma 2.2(1) shows that L_B has p-power order. We first settle the case $p \geq 5$ using the horizontal components of Δ, and then treat $p = 2, 3$ by distinguishing components of coefficient two and one. Let F be the generic fiber of f. It is a regular conic over $K(B)$, and (5.6) gives

$$\deg_{K(B)} \Delta|_F = 2.$$

Suppose first that a horizontal component Γ of Δ is μ-exceptional. The morphism $h\colon \Gamma^\nu \to B$ has degree $d \in \{1, 2\}$. Since $\phi^* L$ is a line bundle on X_1, it is trivial on a neighbourhood of the point $\mu(\Gamma)$. Thus $(\mu^*\phi^* L)|_{\Gamma^\nu} \simeq \mathcal{O}_{\Gamma^\nu}$, and restriction of (5.9) gives $h^* L_B \sim 0$. Lemma 2.1 (1) applied to h yields $dL_B \sim 0$. For $p \geq 3$, coprimality of d and p gives $L_B \sim 0$; for $p = 2$, we obtain $2L_B \sim 0$. We may therefore assume that every horizontal component of Δ is nonexceptional. Such a component has coefficient divisible by $p - 1$. If $p \geq 5$, this contradicts $\deg_{K(B)} \Delta|_F = 2$, completing that case. Hence $p \in \{2, 3\}$, and either $\Delta_{\mathrm{hor}} = 2\Gamma$ with $\Gamma^\nu \simeq B$, or $p = 2$ and Δ_{hor} is reduced.

As in Subsection 2.7, set $r = \dim_{k_1} \operatorname{Der}_k(k_1)$ and $\mathscr{G} = T_{X_1/X}$. Restriction of derivations to k_1 gives an inclusion $\mathscr{G} \hookrightarrow \operatorname{Der}_k(k_1) \otimes_{k_1} \mathcal{O}_{X_1} \simeq \mathcal{O}_{X_1}^{\oplus r}$. Let $\mathscr{G}'$ be the saturation of this image.

Case 1: $\Delta_{\mathrm{hor}} = 2\Gamma$. Let $C = \mu(\Gamma)$, so $C^\nu \simeq B$, and write

$$(p-1)\mathfrak{C} = 2C + C', \qquad C' \geq 0, \qquad C \not\subset \operatorname{Supp} C'.$$

If $(C^2)_{k_1} < 0$, its image $D = \phi(C)_{\mathrm{red}}$ satisfies $(D^2)_k < 0$ by Corollary 2.6. The adjunction formula gives

$$\deg_k K_{D^\nu} + \deg_k \operatorname{Diff}_{D^\nu} = ((K_X + D) \cdot D)_k = (D^2)_k < 0.$$

Thus D^ν has arithmetic genus zero, and $L|_{D^\nu}$ is trivial. Pullback to $\Gamma^\nu \simeq B$ trivializes L_B. Suppose that $(C^2)_{k_1} \geq 0$. By restricting $K_{X_1} + 2C + C' \equiv 0$ to $C^\nu \simeq B$ and applying the adjunction formula, we obtain

$$0 = \deg_{k_1} K_B + \deg_{k_1} \operatorname{Diff}_{C^\nu} + (C^2)_{k_1} + (C' \cdot C)_{k_1}. \tag{5.10}$$

The last three terms are nonnegative, so $\deg_{k_1} K_B \le 0$ and hence $p_a(B) \le 1$. If $p_a(B) = 0$, then $L_B \sim 0$. If $p = 2$ and $p_a(B) = 1$, then Lemma 2.2(2) gives $2L_B \sim 0$.

It remains in this case to consider $p = 3$ and $p_a(B) = 1$. Equation (5.10) gives

$$C \simeq B, \qquad (C^2)_{k_1} = 0, \qquad ((\mathfrak{C} - C) \cdot C)_{k_1} = 0, \tag{5.11}$$

and X_1 is regular along C. In particular, C is nef and $\mathfrak{C} \cdot C = 0$. The curve B is not smooth over k_1, since its geometric genus is zero. We consider separately whether B is geometrically integral or geometrically non-reduced, and rule out both cases.

Subcase 1(a): The base B is geometrically integral. We will show that $\mathfrak{C} \sim \mathfrak{D} + E$ with $E \ge 0$ and $\mathfrak{D} = \mu_* f^* Q$ for a closed point $Q \in B$, contradicting $\mathfrak{C} \cdot C = 0$.

Lemma 5.3 allows us to write $B = \{y^2 z = x^3 + az^3\}$ with $a \in k_1 \setminus k_1^3 = k_1 \setminus k$. Put $U := x/z$ and $Y := y/z$, so that $Y^2 = U^3 + a$ in $K(B) \subseteq K(X_1)$. The zero divisor of Y on B is the unique non-smooth point of B/k_1,

$$Q = \operatorname{Spec}\bigl(k_1[U]/(U^3 + a)\bigr), \qquad [\kappa(Q) : k_1] = 3,$$

with multiplicity one, since Y generates the maximal ideal of $\mathcal{O}_{B,Q}$. Let $\mathfrak{D}$ be the zero divisor of Y on X_1, so $\mathfrak{D} = \mu_* f^* Q$. Since μ is an isomorphism near C and $\Gamma \to B$ is a section,

$$(\mathfrak{D} \cdot C)_{k_1} = (f^* Q \cdot \Gamma)_{k_1} = \deg_{k_1} Q = 3. \tag{5.12}$$

The equation of B forces $\partial(a)$ to vanish along $\mathfrak{D}$ for every local section ∂ of $\mathscr{G}$. Indeed, any local section ∂ of $\mathscr{G}$ is a k-derivation, so in characteristic three

$$\partial(a) = 2Y\partial(Y).$$

At a prime divisor P where Y vanishes, both Y and $\partial(Y)$ lie in $\mathcal{O}_{X_1,P}$, so $v_P(\partial(a)) \ge v_P(Y)$. Lemma 2.11, applied with $D = \mathfrak{D}$, therefore gives $\mathfrak{C} \sim \mathfrak{D} + E$ for some effective divisor E, as required.

Subcase 1(b): The base B is geometrically non-reduced. We will show that $\mathfrak{M} \neq 0$ and use it to construct a second pencil. This construction will yield $K(X) \cap k_1 \supsetneq k$, contradicting the fact that k is algebraically closed in $K(X)$.

Put $\mathbb{K} = K(B)$, $V = \operatorname{Der}_k(k_1)$, $q = \operatorname{rank} \mathscr{G}$ and $s = r - q$. The section Γ gives a $\mathbb{K}$-rational point of the generic fiber F of f, so

$$F \simeq \mathbb{P}^1_{\mathbb{K}} \quad \text{and} \quad K(X_1) = \mathbb{K}(u),$$

for a transcendental element u. Since B is geometrically non-reduced, X_1 is geometrically non-reduced too, and so is X. Theorem 2.10 therefore gives $\mathfrak{M} \neq 0$.

Since C is nef and $\mathfrak{C} \cdot C = 0$, we have

$$\mathfrak{M} \cdot C = \mathfrak{F} \cdot C = 0.$$

Consider the Gauss map from (2.4),

$$\gamma_0 \colon X_1 \dashrightarrow \operatorname{Gr}(q, V) \hookrightarrow \mathbb{P}(\wedge^s V).$$

It records the subspaces $\mathscr{G}'(x) \subset V \otimes_{k_1} \kappa(x)$, and its hyperplanes define the movable system $\mathcal{M} = \gamma_0^* |\mathcal{O}(1)|$ which contains $\mathfrak{M}$. Because $\mathfrak{M} \neq 0$ and $\mathcal{M}$ has no fixed components, γ_0 is nonconstant. Choose a member of $\mathcal{M}$ not containing C. Its intersection with C is zero, so it is disjoint from C. Consequently γ_0 is defined along C.

Resolve the indeterminacy of $\gamma_0 \circ \mu$ by a birational morphism $\rho \colon \widehat{X}_1 \to \widetilde{X}_1$ with $\widehat{X}_1$ regular. Since γ_0 is a morphism along C and μ is an isomorphism near C, we can take ρ to be an isomorphism near Γ. Write

$$\gamma \colon \widehat{X}_1 \longrightarrow \operatorname{Gr}(q, V) \hookrightarrow \mathbb{P}(\wedge^s V), \qquad A = \gamma^* \mathcal{O}(1).$$

Keeping the notation Γ and F for their transforms, we have

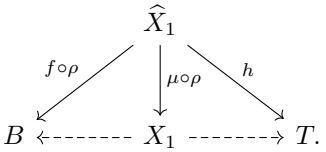
$$(A \cdot \Gamma)_{k_1} = (\Gamma^2)_{k_1} = 0.$$

Since the generic fiber F is unchanged by the resolution of the base points, we have

$$0 \le \deg_{\mathbb{K}}(A|_F) \le \deg_{\mathbb{K}}(((\mu \circ \rho)^* \mathfrak{M})|_F) \le \frac{1}{2} \deg_{\mathbb{K}}(\Delta|_F) = 1.$$

Since $(A \cdot \Gamma)_{k_1} = (\Gamma^2)_{k_1} = 0$ and $\Gamma \not\equiv 0$, the Hodge index theorem gives $(A^2)_{k_1} \le 0$. Then the global generation of A shows $(A^2)_{k_1} = 0$, and the nonconstant map γ has a curve as its image.

Let T be the normalization of this image, and factor γ through $h\colon \widehat{X}_1 \to T$. Then $A = h^*H$, where H is the pullback of $\mathcal{O}(1)$ to T. The two pencils fit into



Since $A \cdot \Gamma = 0$, the image $h(\Gamma)$ is a point, and its residue field embeds into $H^0(\Gamma, \mathcal{O}_\Gamma) = k_1$. Thus T has a k_1-rational point. The regular curve T is smooth at this point, hence generically smooth; as k_1 is separably closed, T is geometrically integral.

If $\deg_{\mathbb{K}}(A|_F) = 0$, then h is constant on F and factors through a nonconstant morphism $B \to T$, contradicting the constancy of $h|_\Gamma$. It follows that $\deg_{\mathbb{K}}(A|_F) = 1$ and the induced morphism

$$h_F\colon \mathbb{P}^1_{\mathbb{K}} \to T_{\mathbb{K}}$$

is finite and satisfies

$$1 = \deg_{\mathbb{K}}(h_F^* H_{\mathbb{K}}) = \deg(h_F)\deg_{k_1} H.$$

Thus $\deg(h_F) = \deg_{k_1} H = 1$. Since H is globally generated of degree one, it induces a isomorphism $T \to \mathbb{P}^1_{k_1}$. If t is its rational coordinate, then

$$K(X_1) = \mathbb{K}(t). \tag{5.13}$$

Let η be the generic point of X_1. Saturation does not change the generic stalk, so $\mathscr{G}_\eta = \mathscr{G}'_\eta$ as subspaces of $V \otimes_{k_1} K(X_1)$. Since γ factors through T and $K(T) = k_1(t)$, this subspace is defined over $k_1(t)$. More precisely, the generic point of T maps to a $k_1(t)$-point of $\mathrm{Gr}(q, V)$, that is, to a q-dimensional subspace

$$W(t) \subset V \otimes_{k_1} k_1(t) \quad \text{with} \quad \mathscr{G}'_\eta = W(t) \otimes_{k_1(t)} K(X_1).$$

Thus, in this product description, the subspace of foliation directions depends only on t.

We now use this description of $\mathscr{G}'_\eta$ to construct an element $a \in k_1 \setminus k$ annihilated by every derivation in $\mathscr{G}$, which will contradict the algebraic closedness of k in $K(X)$. Restricting derivations to k_1 gives an exact sequence of $\mathbb{K}$-vector spaces

$$0 \longrightarrow \mathrm{Der}_{k_1}(\mathbb{K}) \longrightarrow \mathrm{Der}_k(\mathbb{K}) \xrightarrow{\text{res}} \mathrm{Der}_k(k_1, \mathbb{K}) \cong V \otimes_{k_1} \mathbb{K},$$

and we define $\mathfrak{h} \subset V \otimes_{k_1} \mathbb{K}$ to be the image of res. Geometrically, res is the differential $T_{B/k} \to V \otimes_{k_1} \mathcal{O}_B$ of the structure morphism $B \to \operatorname{Spec} k_1$ at the generic point of B. Thus $\mathfrak{h}$ consists of the k-derivations $k_1 \to \mathbb{K}$ that extend to k-derivations of $\mathbb{K}$, that is, the tangent directions of $\operatorname{Spec} k_1$ over k that lift to rational vector fields on B.

Restriction of derivations is dual to the natural map

$$\alpha\colon \mathbb{K} \otimes_{k_1} \Omega_{k_1/k} \to \Omega_{\mathbb{K}/k}$$

So $\mathfrak{h} = \mathrm{Ann}(\ker \alpha)$

Note that $\Omega_{k_1/k} = \Omega_{k_1/\mathbb{F}_3}$ and $\Omega_{\mathbb{K}/k} = \Omega_{\mathbb{K}/\mathbb{F}_3}$, so α is also the natural map of differentials over $\mathbb{F}_3$. The extension $\mathbb{K}/k_1$ is not separably generated, because B is geometrically non-reduced. Hence the differential criterion for separability [SP, Tag 031X] shows that α is not injective. Consequently $\mathfrak{h} \neq V \otimes_{k_1} \mathbb{K}$. We regard V as a k_1-subspace of $V \otimes_{k_1} \mathbb{K}$ via $v \mapsto v \otimes 1$. Under the identification $\mathrm{Der}_k(k_1, \mathbb{K}) \cong V \otimes_{k_1} \mathbb{K}$, this sends $\delta \in \mathrm{Der}_k(k_1)$ to its composite with the inclusion $k_1 \subset \mathbb{K}$. Since V spans $V \otimes_{k_1} \mathbb{K}$ over $\mathbb{K}$, it follows that

$$\mathfrak{d} := V \cap \mathfrak{h} = \{\delta \in \mathrm{Der}_k(k_1) : \delta \text{ extends to a derivation of } \mathbb{K}\}$$

is a proper k_1-subspace of V. If $\delta_1, \delta_2 \in \mathfrak{d}$ extend to derivations D_1, D_2 of $\mathbb{K}$, then $[D_1, D_2]$ and D_1^3 extend $[\delta_1, \delta_2]$ and δ_1^3, so $\mathfrak{d}$ is a restricted Lie subalgebra of V.

Every $D \in \mathscr{G}_\eta$ is a k-derivation of $K(X_1) = \mathbb{K}(t)$, and $D|_{k_1} = (D|_{\mathbb{K}})|_{k_1}$ with $D|_{\mathbb{K}} \in \mathrm{Der}_k(\mathbb{K}, \mathbb{K}(t)) = \mathrm{Der}_k(\mathbb{K}) \otimes_{\mathbb{K}} \mathbb{K}(t)$. Hence $D|_{k_1} \in \mathfrak{h} \otimes_{\mathbb{K}} \mathbb{K}(t)$ for every $D \in \mathscr{G}_\eta$. Since $\mathscr{G}_\eta$ sits in $V \otimes_{k_1} K(X_1)$ via $D \mapsto D|_{k_1}$, this means

$$W(t) \otimes_{k_1(t)} \mathbb{K}(t) = \mathscr{G}_\eta \subseteq \mathfrak{h} \otimes_{\mathbb{K}} \mathbb{K}(t).$$

Let $w \in W(t)$ be arbitrary. After multiplying w by a common denominator of its coordinates in a k_1-basis of V, we may write $w = \sum_i v_i t^i$ with $v_i \in V$. Its image modulo $\mathfrak{h}$ is zero, and transcendence of t over

$\mathbb{K}$ forces each v_i to lie in $\mathfrak{h}$, hence in $\mathfrak{d}$. It follows that $\mathscr{G}_\eta = \mathscr{G}'_\eta \subseteq \mathfrak{d} \otimes_{k_1} \mathbb{K}(t)$. Let $k_1^{\mathfrak{d}}$ be the common constant field of $\mathfrak{d}$. Jacobson's correspondence [Ger64, pp. 561, 565] gives

$$[k_1 : k_1^{\mathfrak{d}}] = 3^{\dim_{k_1} \mathfrak{d}} < 3^r = [k_1 : k].$$

Choose $a \in k_1^{\mathfrak{d}} \setminus k$. Every derivation in $\mathscr{G}$ annihilates a, so $a \in K(X_1)^{\mathscr{G}} = K(X)$, contradicting the algebraic closedness of k in $K(X)$. This finishes the proof of this subcase.

Case 2: $p = 2$ and Δ_{hor} is reduced. Choose a horizontal component Γ of Δ, which has coefficient one. Put $h\colon \Gamma^\nu \to B$ and $e = \deg h \in \{1, 2\}$. The curve Γ is the strict transform of a component C of $\mathfrak{C}$, and we write $\mathfrak{C} = C + C'$ with $C' \geq 0$ not containing C. Adjunction gives

$$0 = \deg_{k_1} K_{C^\nu} + \deg_{k_1} \mathrm{Diff}_{C^\nu} + (C' \cdot C)_{k_1}.$$

Thus C^ν has arithmetic genus at most one over its field of constants. In genus zero, $h^* L_B$ is trivial, so Lemma 2.1 (1) again gives $2L_B \sim 0$. In genus one, we have $\deg_{k_1} \mathrm{Diff}_{C^\nu} = C' \cdot C = 0$, so C is regular, X_1 is regular along C, and C' is disjoint from C. The minimal resolution μ is therefore an isomorphism near C, and we identify Γ with C. If $(C^2)_{k_1} < 0$, the argument as in Case 1 with $D = \phi(C)_{\mathrm{red}}$ trivializes $h^* L_B$, so Lemma 2.1 (1) again gives $2L_B \sim 0$. We may assume $(C^2)_{k_1} \geq 0$, so $(\Gamma^2)_{k_1} \geq 0$ and Γ is nef on $\widetilde{X}_1$.

We distinguish whether the generic conic F is smooth or quasilinear.

Subcase 2(a): The generic conic F is smooth. Put $\Delta_0 := \Delta - \Gamma \geq 0$. Then

$$-(K_{\widetilde{X}_1} + \Delta_0) \equiv \Gamma, \qquad \deg_{K(B)} \Delta_0|_F = 2 - e \in \{0, 1\}.$$

Thus $-(K_{\widetilde{X}_1} + \Delta_0)$ is nef. Since F is smooth, the geometric generic fiber $(F_{\overline{K(B)}}, \Delta_0|_{F_{\overline{K(B)}}})$ of $f\colon (\widetilde{X}_1, \Delta_0) \to B$ is $(\mathbb{P}^1, 0)$ or $(\mathbb{P}^1, P)$ for one reduced point P, hence is F-pure. Lemma 2.9(1) gives $p_a(B) \leq 1$, and Lemma 2.2(2) gives $2L_B \sim 0$.

Subcase 2(b): The generic conic F is quasilinear. The conic F has no $K(B)$-rational point, $e = 2$ and the extension $K(C)/K(B)$ is purely inseparable. Put $\ell = H^0(C, \mathcal{O}_C)$. Since k_1 is separably closed and $e = 2$, either $\ell = k_1$ or $[\ell : k_1] = 2$.

First suppose that $\ell = k_1$. Since $C \to B$ is purely inseparable and B has geometric genus zero, C also has geometric genus zero. As $H^0(C, \mathcal{O}_C) = k_1$, Lemma 2.2 (2) shows that the curve $\widehat{C} = (C \times_{k_1} k_1^{1/2})^\nu_{\mathrm{red}}$ has arithmetic genus zero. The composite $\widehat{C} \to C \to B$ has height one and the pullback of L_B to $\widehat{C}$ is trivial, so Lemma 2.1(2) gives $2L_B \sim 0$.

Finally suppose that $[\ell : k_1] = 2$, and write $\ell = k_1(\sqrt{a})$ with $a \in k_1 \setminus k_1^2 = k_1 \setminus k$. Then $K(C) = K(B)(\sqrt{a})$. Since F is quasilinear, it is defined by a quadratic form

$$sx^2 + ty^2 + rz^2 \quad \text{with } s, t, r \in K(B).$$

Since $p = 2$, we have $(\lambda x + \mu y + \nu z)^2 = \lambda^2 x^2 + \mu^2 y^2 + \nu^2 z^2$, so every linear change of coordinates keeps the form a combination of x^2, y^2, z^2 only. After a linear change of coordinates, we may assume that $\{z = 0\}$ is the line spanned by the degree-two point $C|_F$. Then $C|_F$ is the zero of $sx^2 + ty^2$ on this line, so $K(B)(\sqrt{t/s}) = K(C) = K(B)(\sqrt{a})$. Hence $t/s = \alpha^2 + \beta^2 a$ for some $\alpha, \beta \in K(B)$, and a linear change of x and y gives

$$F = \{x^2 + ay^2 + bz^2 = 0\}, \qquad b \in K(B), \qquad C|_F = \{z = 0\}.$$

On $F \cap \{y \neq 0\}$, put $u = x/y$ and $v = z/y$. At the generic point of C, the function v is a uniformizer, and b, viewed as a function on X_1 via f, is regular there since C dominates B. For a local section θ of $\mathscr{G} = T_{X_1/X}$, differentiating $u^2 + a + bv^2 = 0$ gives

$$\theta(a) = v^2 \theta(b), \tag{5.14}$$

where $\theta(b)$ is regular along C. Thus $\theta(a)$ vanishes to order at least two along C, and Lemma 2.11, applied with $D = 2C$, gives $\mathfrak{C} \sim 2C + E$ for some $E \geq 0$. This contradicts $\deg_{K(B)}((\mu^* \mathfrak{C})|_F) \leq \deg_{K(B)}(\Delta|_F) = 2$. This finishes the proof of this subcase and hence the whole proof. □

5.3 The case $-K_X$ nef and $K_X \not\equiv 0$

Proposition 5.5. *Let k be a field of characteristic $p > 0$, and let X be a regular, geometrically non-normal projective k-surface such that $H^0(X, \mathcal{O}_X) = k$, $-K_X$ is nef and $K_X \not\equiv 0$, and $\dim \mathrm{Alb}_{X/k} = 0$. Then every numerically trivial line bundle L on X satisfies the following:*

(1) *If $p \geq 5$, then $L \sim 0$.*

(2) *If* $p = 3$, *then* $3L \sim 0$.
(3) *If* $p = 2$, *then* $2^2 L \sim 0$.

Proof. We may assume that k is F-finite and separably closed. Since $-K_X$ is nef and numerically nontrivial, the K_X-MMP ends with a Mori fiber space

$$\begin{array}{ccc} X & \xrightarrow{\epsilon} & Z \\ & \searrow^{f} & \downarrow g \\ & & B\,. \end{array}$$

The divisor $-K_Z$ is nef.

Suppose first that $\dim B = 0$. By Lemma 2.21 (1), there exists a numerically trivial line bundle L_Z on the regular del Pezzo surface Z such that $L \simeq \epsilon^* L_Z$. Then the conclusion follows from Proposition 2.13.

Suppose that $\dim B = 1$. By Lemma 2.21(2), there exists a numerically trivial line bundle L_B on B such that $L \simeq f^* L_B$. Since $\mathrm{Alb}_{X/k}$ is trivial, the curve B has geometric genus zero.

We divide the argument into the cases $p > 2$ and $p = 2$.

Case (a): $p > 2$. By [BT22, Proposition 2.18], the Mori fibration g is smooth because $p > 2$ and k is separably closed. Its geometric generic fiber is $\mathbb{P}^1$, so positivity (Lemma 2.9 (1)) gives $p_a(B) \leq 1$. If $p \geq 5$, Lemma 2.2 (2) shows $p_a(B) = 0$ and hence $L_B \sim 0$. If $p = 3$, the same lemma gives $3L_B \sim 0$. Pulling back by f proves the required bounds for L.

Case (b): $p = 2$. If $K_Z^2 > 0$, then $-K_Z$ is nef and big, so Proposition 2.13 applied to $g^* L_B$ gives $4L \sim 0$. We may therefore assume that $K_Z^2 = 0$. If the generic conic of g is smooth, Lemma 2.9 (1) gives $p_a(B) \leq 1$, and Lemma 2.2 (2) gives $2L_B \sim 0$. Hence we may assume that the generic conic is quasilinear.

Set $k_1 = k^{1/2}$ and

$$B_0 := \bigl(B \times_k k_1\bigr)^{\nu}_{\mathrm{red}}, \qquad Y := \bigl(Z \times_k k_1\bigr)^{\nu}_{\mathrm{red}},$$

and let $\phi\colon Y \to Z$ be the natural morphism. We have $H^0(Y, \mathcal{O}_Y) = H^0(B_0, \mathcal{O}_{B_0}) = k_1$. Let

$$Y \xrightarrow{h} B_1 \xrightarrow{\tau} B$$

be the Stein factorization of $g \circ \phi$ and put $M := \tau^* L_B$. The morphism τ has height at most one, so it suffices to prove $2M \sim 0$. The curve B_1 has geometric genus zero and $H^0(B_1, \mathcal{O}_{B_1}) = k_1$. Since Y is normal and dominates $(B \times_k k_1)_{\mathrm{red}}$, the morphism $g \circ \phi$ factors as $Y \to B_0 \xrightarrow{\beta} B$. The Stein factorization of $Y \to B_0$ is then also a Stein factorization of $g \circ \phi$, so $\tau = \beta \circ \tau_0$ for a finite morphism $\tau_0\colon B_1 \to B_0$.

Let $\mu\colon S \to Y$ be a minimal resolution and set

$$s := h \circ \mu, \qquad N_Y := \phi^*(-K_Z), \qquad N := \mu^* N_Y.$$

The morphisms introduced so far fit into the commutative diagram

$$\begin{array}{ccccccc} S & \xrightarrow{\mu} & Y & & \xrightarrow{\phi} & & Z & \xleftarrow{\epsilon} & X \\ & \searrow^{s} & \downarrow h & & & & \downarrow g & \swarrow_{f} & \\ & & B_1 & \xrightarrow{\tau_0} & B_0 & \xrightarrow{\beta} & B\,. & & \\ & & & \searrow & & \nearrow & & & \\ & & & & \tau & & & & \end{array}$$

The canonical base-change formula and the minimal-resolution argument in (2.7) give

$$K_S + \Delta \sim -N$$

for an effective integral divisor Δ. The divisor N is nef with $N^2 = 0$, and N_Y is h-ample.

Put $\mathbb{K} := K(B)$ and $\mathbb{K}_0 := K(B_0)$. The generic conic V of g is quasilinear, so it is defined by an equation

$$c_1 x^2 + c_2 y^2 + c_3 z^2 = 0 \quad \text{with } c_1, c_2, c_3 \in \mathbb{K}. \tag{5.15}$$

Suppose first that $V_{\mathbb{K}_0}$ is not regular. After a linear change of coordinates, its equation is $x^2 = 0$ or $x^2 + ay^2 = 0$ with $a \notin \mathbb{K}_0^2$. Its reduced normalization is therefore $\mathbb{P}^1$ over $\mathbb{K}_0$ or $\mathbb{K}_0(\sqrt{a})$, respectively, and the hyperplane bundle pulls back to $\mathcal{O}_{\mathbb{P}^1}(1)$. Thus the generic fiber F of s is $\mathbb{P}^1_{K(B_1)}$, and

$$\deg N|_F = 1, \qquad \deg \Delta|_F = 1.$$

Consequently, the geometric generic fiber $(F_{\overline{K(B_1)}}, \Delta|_{F_{\overline{K(B_1)}}})$ of $s\colon (S,\Delta) \to B_1$ is $(\mathbb{P}^1, P)$ for one reduced point P, hence is F-pure. Lemma 2.9 (1) gives $p_a(B_1) \le 1$, and Lemma 2.2 (2) gives $2M \sim 0$.

It remains to treat the case where $V_{\mathbb{K}_0}$ is regular. Then $H^0(V_{\mathbb{K}_0}, \mathcal{O}_{V_{\mathbb{K}_0}}) = \mathbb{K}_0$, so $B_1 = B_0$. We next show that B is not smooth. Suppose to the contrary that B is smooth. Then $\mathbb{K}$ is separable over $k(t)$ for some $t \in \mathbb{K}$, so $\mathbb{K} = \mathbb{K}^2k(t)$ and $t \notin \mathbb{K}^2k$, hence $[\mathbb{K} : \mathbb{K}^2k] = 2$. Consequently, c_1, c_2, c_3 are linearly dependent over $\mathbb{K}^2k = \mathbb{K}_0^2$, so $V_{\mathbb{K}_0}$ has a $\mathbb{K}_0$-rational point. By [Tan21, Theorem 9.10 (3)], the regular conic $V_{\mathbb{K}_0}$ is then isomorphic to $\mathbb{P}^1_{\mathbb{K}_0}$ and hence smooth, which is impossible because all partial derivatives of the polynomial in (5.15) vanish. Hence B is not smooth.

By Theorem 2.10 applied to B, there is an integral divisor $D_B > 0$ on $B_1 = B_0$ such that

$$K_{B_1} + D_B \sim \tau^* K_B.$$

The conic fibration g is flat and Gorenstein, so $W := Z \times_B B_1$ is Gorenstein and is integral because its generic fiber is integral. Its normalization is Y, since both have function field $K(Z)k_1$. Relative dualizing base change and the conductor formula therefore give

$$K_Y + C_{\mathrm{nor}} + h^* D_B \sim \phi^* K_Z, \qquad C_{\mathrm{nor}} \ge 0.$$

Since $K_Y + C_{\mathrm{nor}} \sim \phi^* K_Z - h^* D_B$ is Cartier and μ is the minimal resolution, we have $\mu^*(K_Y + C_{\mathrm{nor}}) \sim K_S + \Delta'$ with $\Delta' \ge 0$ integral, so we may take $\Delta := \Delta' + s^* D_B$, and then

$$K_S + \Delta \sim -N, \qquad \Delta \ge s^* D_B.$$

In particular, $\Delta \cdot N > 0$ because N_Y is h-ample, and hence $K_S \cdot N < 0$.

Since $(K_S - mN) \cdot N < 0$, we have $h^2(S, mN) = 0$, so Riemann–Roch gives

$$h^0(S, mN) \ge \chi(S, \mathcal{O}_S(mN)) = \chi(S, \mathcal{O}_S) - \frac{m}{2} K_S \cdot N \to \infty.$$

For m sufficiently large, the moving part A of $|mN|$ satisfies $N \cdot A = 0$. The Hodge index theorem and movability give $A^2 = 0$, so $|A|$ has no base points. Its Stein factorization defines a morphism $q\colon S \to D$ to a regular projective curve, with N numerically proportional to a fiber of q. The generic fiber C is regular with $H^0(C, \mathcal{O}_C) = K(D)$, and

$$\deg K_C = -\deg \Delta|_C < 0.$$

Thus $p_a(C) = 0$ and $\deg \Delta|_C = 2$. Running a K_S-MMP over D and applying Lemma 2.21(2), we obtain a numerically trivial line bundle M_D on D such that

$$s^* M \simeq q^* M_D.$$

Since $(s^* D_B) \cdot N > 0$ and N is numerically proportional to a fiber of q, the divisor $s^* D_B$ has a q-horizontal prime component Γ. The morphisms used below fit into the commutative diagram

$$\begin{array}{ccccc} \Gamma^\nu & \longrightarrow & S & \xrightarrow{\ s\ } & B_1\,, \\ & \searrow & \downarrow{\scriptstyle q} & & \\ & & D & & \end{array}$$

where $\Gamma^\nu \to S$ is the normalization of Γ followed by the inclusion. Since $\Delta \ge s^* D_B$, the finite morphism $\Gamma^\nu \to D$ has degree

$$1 \le d = \deg_{K(D)}(\Gamma|_C) \le \deg_{K(D)}(\Delta|_C) = 2.$$

The curve Γ is s-vertical, so $s^* M|_{\Gamma^\nu}$ is trivial, being pulled back from a closed point of B_1. Restricting $s^* M \simeq q^* M_D$ to Γ^ν and applying Lemma 2.1 (1) gives $d M_D \sim 0$, hence $2M_D \sim 0$. Since $s_* \mathcal{O}_S = \mathcal{O}_{B_1}$, pullback is injective and $2M \sim 0$. This finishes the proof of this subcase and thus the whole proof. □

6 Twisted covers and triviality in large characteristic

In this section, we prove that every numerically trivial line bundle on a surface as in Theorem 1.1(C) is trivial when $p \ge 7$.

6.1 The t-twisted μ_p-cover

Let k be a field of characteristic $p > 0$, and let X be a k-scheme. Let L be a line bundle on X such that $L^{\otimes p} \simeq \mathcal{O}_X$. We fix an isomorphism

$$\tau\colon L^{\otimes p} \xrightarrow{\sim} \mathcal{O}_X.$$

Let t be transcendental over k, set $K = k(t)$, and write $X_K = X \times_k K$. Denote the pullbacks of L and τ to X_K by L_K and τ_K, respectively, and define the trivialization

$$\tau_t\colon L_K^{\otimes p} \xrightarrow{\sim} \mathcal{O}_{X_K}, \qquad v \longmapsto t^{-1}\tau_K(v).$$

Regarding τ_t as a nowhere vanishing section of $L_K^{-p} = \mathcal{H}om(L_K^{\otimes p}, \mathcal{O}_{X_K})$, let $\lambda_t\colon L_K^{-p} \xrightarrow{\sim} \mathcal{O}_{X_K}$ be the isomorphism with $\lambda_t(\tau_t) = 1$. We define an $\mathcal{O}_{X_K}$-algebra

$$\mathcal{A}_{L,t} := \bigoplus_{i=0}^{p-1} L_K^{-i}$$

by taking the multiplication to be the usual tensor product on local sections of degrees i and j with $i + j < p$, and the map

$$L_K^{-i} \otimes L_K^{-j} \simeq L_K^{-(i+j-p)} \otimes L_K^{-p} \xrightarrow{1\otimes\lambda_t} L_K^{-(i+j-p)}$$

when $i + j \geq p$. Let

$$\pi\colon \widehat{Y} := \operatorname{Spec}_{X_K} \mathcal{A}_{L,t} \longrightarrow X_K.$$

The $\mathbb{Z}/p\mathbb{Z}$-grading induces a μ_p-action on $\widehat{Y}$, and π is a μ_p-torsor. In particular, it is finite flat of degree p. We call π the *t-twisted μ_p-cover* associated with (L, τ).

To describe it locally, let $U = \operatorname{Spec} R \subset X$ be an open subset on which L is free, and choose a basis e of $L|_U$. Its dual basis is the section

$$z = e^\vee \in H^0(U, L^{-1}), \qquad z(e) = 1.$$

Thus $z(ae) = a$ for every local function a, and $z^{\otimes i}$ is a basis of $L^{-i}|_U$. Write

$$\tau(e^{\otimes p}) = u \in R^\times.$$

After passing to U_K, we have $\tau_t(e^{\otimes p}) = t^{-1}u$, that is,

$$\tau_t = t^{-1}uz^{\otimes p}$$

as a section of L_K^{-p}. Consequently, $\lambda_t(z^{\otimes p}) = tu^{-1}$, so the degree-one element $z \in L_K^{-1}$ satisfies $z^p = tu^{-1}$ in the covering algebra. Hence the cover over U_K is

$$\operatorname{Spec}\big((R \otimes_k K)[z]/(z^p - tu^{-1})\big).$$

6.2 Triviality in characteristic $p \geq 7$

Proposition 6.1. *Let k be a field of characteristic $p > 0$, and let X be a regular projective k-variety such that $H^0(X, \mathcal{O}_X) = k$. Let L be a line bundle on X of exact order p, and let*

$$\pi\colon \widehat{Y} = \operatorname{Spec}_{X_K} \mathcal{A}_{L,t} \longrightarrow X_K$$

be the t-twisted μ_p-cover associated with (L, τ) for an isomorphism $\tau\colon L^{\otimes p} \xrightarrow{\sim} \mathcal{O}_X$. Then

(1) *$\widehat{Y}$ is a regular projective variety over K with $H^0(\widehat{Y}, \mathcal{O}_{\widehat{Y}}) = K$,*
(2) *$\omega_{\widehat{Y}} \simeq \pi^*\omega_{X_K}$ and $\pi^*L_K \simeq \mathcal{O}_{\widehat{Y}}$.*

Assume furthermore that $X_{\overline{k}}$ is integral and $\nu^*L_{\overline{k}} \simeq \mathcal{O}_Z$, where $\nu\colon Z = X_{\overline{k}}^{\nu} \to X_{\overline{k}}$ is the normalization. Then

(3) *$\widehat{Y}$ is geometrically non-reduced over K.*

Proof. (1) The purely transcendental extension $k \subset K$ is separable, so X_K is regular. Let $U = \operatorname{Spec} R \subset X$ be an affine open on which L is free, and let e, $z = e^\vee$, and $u = \tau(e^{\otimes p}) \in R^\times$ be as in Subsection 6.1. By the local description,

$$\Gamma(U_K, \mathcal{A}_{L,t}) = \bigoplus_{i=0}^{p-1} (R \otimes_k K)z^i \simeq (R \otimes_k K)[z]/(z^p - tu^{-1}).$$

Since $R \otimes_k K$ is the localization of $R[t]$ at $k[t] \setminus \{0\}$, the coordinate ring $(R \otimes_k K)[z]/(z^p - tu^{-1})$ of $\pi^{-1}(U_K)$ is a localization of $R[t,z]/(z^p - tu^{-1})$. The latter ring is isomorphic to $R[z]$ via $t \mapsto uz^p$, which is regular since U is open in the regular variety X, so $\pi^{-1}(U_K)$ is regular. These inverse images cover $\widehat{Y}$, proving its regularity. Finally, π is finite, hence projective, and X_K is projective over K. Their composition shows that $\widehat{Y}$ is projective over K.

Next, for $1 \le i \le p-1$, the line bundle L^{-i} is nontrivial and $(L^{-i})^{\otimes p} \simeq \mathcal{O}_X$, so $H^0(X, L^{-i}) = 0$. Therefore $H^0(X_K, L_K^{-i}) = 0$. Since π is affine and $\pi_* \mathcal{O}_{\widehat{Y}} = \mathcal{A}_{L,t}$, we obtain

$$H^0(\widehat{Y}, \mathcal{O}_{\widehat{Y}}) = \bigoplus_{i=0}^{p-1} H^0(X_K, L_K^{-i}) = K.$$

Thus $\widehat{Y}$ is connected, hence integral because it is regular.

(2) We first compute the canonical sheaf of $\widehat{Y}$. Multiplication followed by projection onto the degree-zero summand defines an $\mathcal{O}_{X_K}$-bilinear pairing

$$\mathcal{A}_{L,t} \otimes \mathcal{A}_{L,t} \longrightarrow \mathcal{O}_{X_K}, \qquad a \otimes b \longmapsto \mathrm{pr}_0(ab).$$

This pairing is perfect, because $\lambda_t \colon L_K^{-i} \otimes L_K^{-(p-i)} = L_K^{-p} \to \mathcal{O}_{X_K}$ is an isomorphism for $1 \le i < p$. Therefore the induced map

$$\mathcal{A}_{L,t} \longrightarrow \mathcal{H}om_{X_K}(\mathcal{A}_{L,t}, \mathcal{O}_{X_K}), \qquad a \longmapsto \big(b \mapsto \mathrm{pr}_0(ab)\big),$$

is an isomorphism of $\mathcal{A}_{L,t}$-modules, where $\mathcal{A}_{L,t}$ acts on the target by precomposition with multiplication. Since $\pi \colon \widehat{Y} \to X_K$ is a finite flat morphism of regular varieties, duality for finite morphisms gives

$$\pi_* \omega_{\widehat{Y}} \simeq \mathcal{H}om_{X_K}(\mathcal{A}_{L,t}, \omega_{X_K}).$$

The algebra $\mathcal{A}_{L,t}$ is locally free over $\mathcal{O}_{X_K}$, so its self-duality gives

$$\begin{aligned} \mathcal{H}om_{X_K}(\mathcal{A}_{L,t}, \omega_{X_K}) &\simeq \mathcal{H}om_{X_K}(\mathcal{A}_{L,t}, \mathcal{O}_{X_K}) \otimes \omega_{X_K} \\ &\simeq \mathcal{A}_{L,t} \otimes \omega_{X_K} \simeq \pi_* \pi^* \omega_{X_K}. \end{aligned}$$

These are isomorphisms of $\mathcal{A}_{L,t}$-modules. Since π_* identifies quasi-coherent $\mathcal{O}_{\widehat{Y}}$-modules with quasi-coherent $\mathcal{A}_{L,t}$-modules, we obtain

$$\omega_{\widehat{Y}} \simeq \pi^* \omega_{X_K}.$$

Next, we trivialize $\pi^* L_K$. Under the adjunction $\mathrm{Hom}_{\mathcal{O}_{\widehat{Y}}}(\pi^* L_K^{-1}, \mathcal{O}_{\widehat{Y}}) = \mathrm{Hom}_{\mathcal{O}_{X_K}}(L_K^{-1}, \pi_* \mathcal{O}_{\widehat{Y}})$, the inclusion of the degree-one summand $L_K^{-1} \hookrightarrow \mathcal{A}_{L,t} = \pi_* \mathcal{O}_{\widehat{Y}}$ corresponds to the homomorphism

$$\pi^* L_K^{-1} \longrightarrow \mathcal{O}_{\widehat{Y}}, \qquad a \otimes w \longmapsto aw, \tag{6.1}$$

where a is a local function on $\widehat{Y}$ and w is a local section of L_K^{-1}, viewed as a function on $\widehat{Y}$. Over $\pi^{-1}(U_K)$, the line bundle $\pi^* L_K^{-1}$ has the basis $1 \otimes z$, where $z = e^\vee$, and this map sends it to the element z of the covering ring $(R \otimes_k K)[z]/(z^p - tu^{-1})$. The relation $z^p = tu^{-1}$ shows that z is a unit, so (6.1) is an isomorphism over $\pi^{-1}(U_K)$. These affine opens cover $\widehat{Y}$, so $\pi^* L_K^{-1} \simeq \mathcal{O}_{\widehat{Y}}$. Dualizing gives $\pi^* L_K \simeq \mathcal{O}_{\widehat{Y}}$.

(3) Let Ω be an algebraic closure of K, choose a k-embedding $\overline{k} \hookrightarrow \Omega$, and identify $\overline{k}$ with its image. By assumption, X_Ω is integral, and the base change $\nu_\Omega \colon Z_\Omega \to X_\Omega$ is its normalization.

Fix an isomorphism $\alpha \colon \nu^* L_{\overline{k}} \xrightarrow{\sim} \mathcal{O}_Z$, and denote its base change by $\alpha_\Omega \colon L_\Omega|_{Z_\Omega} \xrightarrow{\sim} \mathcal{O}_{Z_\Omega}$. Set $s_Z := \alpha_\Omega^{-1}(1)$. Let $\tau_Z \colon (L_\Omega|_{Z_\Omega})^{\otimes p} \xrightarrow{\sim} \mathcal{O}_{Z_\Omega}$ be the pullback of τ_t to Z_Ω, and set

$$c := \tau_Z(s_Z^{\otimes p}) \in H^0(Z_\Omega, \mathcal{O}_{Z_\Omega})^\times = \Omega^\times.$$

Let $F = \Omega(Z_\Omega) = \Omega(X_\Omega)$ be the common function field. At the generic point, the element $w = s_Z^\vee$ satisfies $w^p = c^{-1}$, so the generic covering algebra is

$$F[w]/(w^p - c^{-1}).$$

Since Ω is algebraically closed, we have $c^{-1} = a^p$ for some $a \in \Omega^\times$, so the generic covering algebra $F[w]/((w-a)^p)$ is non-reduced. Hence $(\widehat{Y})_\Omega$ is non-reduced, that is, $\widehat{Y}$ is geometrically non-reduced over K. $\square$

Corollary 6.2. *Let k be a field of characteristic $p \geq 7$, and let X be a regular, geometrically non-normal projective k-surface such that $H^0(X, \mathcal{O}_X) = k$, $K_X \equiv 0$. Then every numerically trivial line bundle L on X is trivial.*

Proof. We may assume that k is separably closed. Since $-K_X$ is nef and $p \geq 7$, [WZ26, Theorem 1.2] shows that X is geometrically integral. We have $\dim \mathrm{Alb}_{X/k} = 0$ by Proposition 4.1 since $p \geq 5$. It follows from Proposition 5.4 that $L^{\otimes p} \simeq \mathcal{O}_X$. Moreover, (5.7) in the proof of Proposition 5.4 gives $\phi^* L \simeq \mathcal{O}_Z$ for the natural morphism $\phi\colon Z := X_k^\nu \to X$. If L were nontrivial, it would have exact order p. Proposition 6.1 would then produce a regular projective surface $\widehat{Y}$ over $K = k(t)$ such that $H^0(\widehat{Y}, \mathcal{O}_{\widehat{Y}}) = K$ and $K_{\widehat{Y}} \equiv \pi^* K_{X_K} \equiv 0$, while $\widehat{Y}$ is geometrically non-reduced. This contradicts [WZ26, Theorem 1.2]. Hence $L \simeq \mathcal{O}_X$. □

7 Examples

7.1 Diagonal quotients in characteristics two and three

By Proposition 4.1 (4), every surface in Case (B) is essentially a diagonal quotient $(E \times C)/G$, where E is the Albanese variety, C is an Albanese fiber, and $G \subset E$ is a finite subgroup scheme. Table 1 lists four such examples $X = (C \times_k E)/G$. They show that the bounds 12 and 6 in Case (B) are optimal for $p = 2$ and $p = 3$, respectively.

Example	p	k	C	G	$\mathrm{ord}(K_X)$
7.4	2	$\mathbf{F}_2(a,b)$	$y^2 = ax^4 + bz^4$	μ_4, non-reduced	4
7.5	2	$\mathbf{F}_2(s)$	$y^2 z = x^3 + sz^3$	μ_3, étale	3
7.6	3	$\mathbf{F}_3(s)$	$y^2 z = x^3 + sz^3$	μ_2, étale	2
7.7	3	$\mathbf{F}_3(s,t)$	$sx^3 + ty^3 + z^3 = 0$	μ_3, non-reduced	3

Table 1: Examples in Case (B); $C \subset \mathbb{P}^2_k$ except in Example 7.4, where $C \subset \mathbb{P}_k(1,2,1)$.

Lemma 7.1. *Let G be a finite k-group scheme that is commutative, étale, or linearly reductive, and let $q\colon Y \to X$ be a G-torsor between projective Gorenstein k-schemes such that $H^0(Y, \mathcal{O}_Y) = k$ and $\omega_Y \simeq \mathcal{O}_Y$. Then G acts on the one-dimensional k-vector space $H^0(Y, \omega_Y)$ through a character $\chi\colon G \to \mathbb{G}_m$, and*

$$\mathrm{ord}(K_X) = \mathrm{ord}(\chi),$$

where $\mathrm{ord}(\chi)$ is the least positive integer m such that $\chi^m = 1$.

Proof. Fix a generator η of ω_Y, so that $H^0(Y, \omega_Y) = k\eta$, and put $Y_T := Y \times_k T$ for a k-scheme T. For every T and every $\lambda \in G(T)$, the pullback $\lambda^*\eta$ is again a generator of $\omega_{Y_T/T}$, so $\lambda^*\eta = \chi(\lambda)\eta$ for some $\chi(\lambda) \in H^0(Y_T, \mathcal{O}_{Y_T}^\times) = \Gamma(T, \mathcal{O}_T)^\times$. This assignment is functorial in T and satisfies $\chi(\lambda\mu) = \chi(\lambda)\chi(\mu)$, so it defines a character $\chi\colon G \to \mathbb{G}_m$.

By fpqc descent along the G-torsor q, a line bundle on X is the same as a G-equivariant line bundle on Y. We first show that $q^*\omega_X \simeq \omega_Y$ as G-equivariant line bundles. By Grothendieck duality [SP, Tag 0C10], the canonical isomorphism $\omega_Y \simeq q^*\omega_X \otimes \omega_{Y/X}$ is G-equivariant. Thus it suffices to show that $\omega_{Y/X}$ has a G-invariant generator. We only treat the case $G = \mu_n$, which suffices for the applications below, and refer to [SP, Tags 0FWI and 0ATX] if G is étale, to [Bri, Lemma 4.3.1(ii)] if G is linearly reductive, and to [Via26, §§4.1.4–4.1.5] if G is commutative.

Suppose $G = \mu_n$, and write $q_*\mathcal{O}_Y = \bigoplus_{i \in \mathbb{Z}/n\mathbb{Z}} \mathcal{L}_i$ for the grading, where $\mathcal{L}_0 = \mathcal{O}_X$. Let $\varepsilon\colon q_*\mathcal{O}_Y \to \mathcal{O}_X$ be the projection onto degree zero. As in the proof of Proposition 6.1 (2), the torsor grading makes the pairing $(b, c) \mapsto \varepsilon(bc)$ perfect. By duality, $q_*\omega_{Y/X} \simeq \mathcal{H}om_{\mathcal{O}_X}(q_*\mathcal{O}_Y, \mathcal{O}_X)$ as $q_*\mathcal{O}_Y$-modules, where $(b \cdot \varphi)(c) = \varphi(bc)$. Then the $q_*\mathcal{O}_Y$-linear map $q_*\mathcal{O}_Y \to q_*\omega_{Y/X}$ sending 1 to ε is an isomorphism, so ε generates $\omega_{Y/X}$. Write ε_T for the base change of ε to a k-scheme T. For every $\lambda \in G(T)$ and every local section $\sum_i b_i$ of the base-changed graded algebra, where b_i has degree i, we have $\lambda^* b_i = \lambda^i b_i$ and hence

$$\varepsilon_T\Big(\lambda^* \sum_i b_i\Big) = \varepsilon_T\Big(\sum_i \lambda^i b_i\Big) = b_0 = \varepsilon_T\Big(\sum_i b_i\Big).$$

Thus ε is a G-invariant generator of $\omega_{Y/X}$, so $q^*\omega_X \simeq \omega_Y$ as G-equivariant line bundles.

Consequently, for every $m \geq 1$, the line bundle $\omega_X^{\otimes m}$ is trivial if and only if $\omega_Y^{\otimes m} \simeq \mathcal{O}_Y$ as G-equivariant line bundles, that is, if and only if $\omega_Y^{\otimes m}$ has a G-invariant generator. Every generator of $\omega_Y^{\otimes m}$ is a constant multiple of $\eta^{\otimes m}$, so such an invariant generator exists if and only if $\chi^m = 1$. This proves the assertion. □

For a scheme Y over a field k and a point $y \in Y(k)$, write $y_S \colon S \to Y_S$ for the induced section over a k-scheme S. The *stabilizer* $\mathrm{Stab}(y) \subset \mathrm{Aut}_k(Y)$ is the subgroup functor defined by

$$\mathrm{Stab}(y)(S) := \{\varphi \in \mathrm{Aut}_S(Y_S) \mid \varphi \circ y_S = y_S\}.$$

Given an action $\alpha \colon H \to \mathrm{Aut}_k(Y)$ of a k-group scheme H, the *fixed point functor* $Y^H \subset Y$ is defined on each k-scheme S by

$$Y^H(S) := \{x \in Y(S) \mid h \cdot x_{S'} = x_{S'} \text{ for every } S' \to S \text{ and every } h \in H(S')\},$$

where $x_{S'}$ denotes the pullback of x to S'. The action fixes y scheme-theoretically if $y \in Y^H(k)$, or equivalently, if $\alpha(H) \subseteq \mathrm{Stab}(y)$; see [Bri17, Definition 2.2.5 and Theorem 2.2.6].

Lemma 7.2 (Regular geometrically non-normal diagonal quotients)**.** *Let C be a regular curve over k. Let $G := \mu_n$ act on C, and let $E \to B$ be a G-torsor with E a smooth curve over k. Then the quotient $X := (C \times_k E)/G$ exists as a k-scheme. Write $\alpha \colon G_{\bar{k}} \to \mathrm{Aut}(C_{\bar{k}})$ for the induced action.*

(1) *The scheme X is regular. If C is geometrically reduced, then X is geometrically reduced.*

(2) *Assume that C is geometrically reduced and that $C_{\bar{k}}$ has a unique singular point c. If*

$$\alpha(G_{\bar{k}}) \subseteq \mathrm{Stab}(c),$$

then X is geometrically non-normal.

Proof. The diagonal G-action is free because $E \to B$ is a G-torsor. By [SP, Tag 07S7], the quotient X exists as a scheme and the quotient map $q \colon C \times_k E \to X$ is a G-torsor, hence finite and faithfully flat. Since C is regular and E is smooth over k, the product $C \times_k E$ is regular. Since q is faithfully flat, X is regular by descent. If C is geometrically reduced, then $C_{\bar{k}} \times_{\bar{k}} E_{\bar{k}}$ is reduced, and reducedness descends to $X_{\bar{k}}$. This proves (1).

For (2), work over $\bar{k}$ and write $\overline{C} := C_{\bar{k}}$, $\overline{E} := E_{\bar{k}}$, $\overline{B} := B_{\bar{k}}$, $\overline{G} := G_{\bar{k}}$, and $\overline{X} := X_{\bar{k}}$. The fixed point c gives a $\overline{G}$-equivariant section $\overline{E} \to \overline{C} \times_{\bar{k}} \overline{E}$, which descends to a section

$$s \colon \overline{B} \longrightarrow \overline{X}$$

of the natural morphism $f \colon \overline{X} \to \overline{B}$. The curve $\overline{B}$ is smooth, since geometric regularity descends through the torsor $\overline{E} \to \overline{B}$ by [SP, Tag 07NH]. The two projections induce the following diagram:

$$\begin{array}{ccccc} \overline{C} \times_{\bar{k}} \overline{E} & \xrightarrow{q_{\bar{k}}} & \overline{X} & \longrightarrow & \overline{C}/\overline{G} \\ & & {\scriptstyle f}\downarrow \uparrow {\scriptstyle s} & & \\ & & \overline{B} & & \end{array}$$

Let η be the generic point of a component of $\overline{B}$, put $\ell := \kappa(\eta)$, and let $\xi := s(\eta) \in \overline{X}_\eta(\ell)$. Let Ω be an algebraic closure of ℓ and choose a point $e \in \overline{E}_\eta(\Omega)$. The point e trivializes the torsor $(\overline{E}_\eta)_\Omega$, so the map $x \mapsto (x, e)$ followed by the quotient map gives an isomorphism $\overline{C}_\Omega \simeq (\overline{X}_\eta)_\Omega$. By the construction of s, this isomorphism sends c_Ω to ξ_Ω. Thus $\overline{X}_\eta$ is not smooth at ξ. Since ξ is ℓ-rational, its local ring is not regular by [SP, Tag 00TV]. So $\overline{X}$ is not normal at ξ. □

Remark 7.3. Under the assumptions on C in Lemma 7.2(2), every element of $G_{\bar{k}}(\bar{k})$ fixes c, since automorphisms preserve singular points and c is the only singular point. The stabilizer condition requires the orbit morphism $G_{\bar{k}} \to C_{\bar{k}}$, $g \mapsto g \cdot c$, to be constant with value c as a morphism of schemes. If G is étale, then $G_{\bar{k}}$ is a disjoint union of copies of $\operatorname{Spec} \bar{k}$, so fixing c on geometric points is enough to verify this condition. Thus, when G is étale, Lemma 7.2(2) applies as soon as C is geometrically reduced and $C_{\bar{k}}$ has a unique singular point. For non-reduced G, the orbit morphism may be nonconstant even though every geometric point maps to c, so the stabilizer condition must be checked separately.

Example 7.4 ($p = 2$ and $\mathrm{ord}(K_X) = 4$)**.** Put $k := \mathbf{F}_2(a, b)$, where a, b are algebraically independent over $\mathbf{F}_2$. Let $C := V(y^2 - ax^4 - bz^4) \subset \mathbb{P}_k(1, 2, 1)$. The curve C is regular and integral, with $H^0(C, \mathcal{O}_C) = k$, $p_a(C) = 1$, and $\omega_C \simeq \mathcal{O}_C$. Let μ_4 act on C by

$$\lambda \cdot [x : y : z] = [\lambda x : y : z], \qquad \lambda^4 = 1.$$

On $U := C \cap \{z \neq 0\}$, put $\xi := x/z$, $\eta := y/z^2$, $f := \eta^2 - a\xi^4 - b$, and $\mathcal{I} := (f)$. By adjunction,

$$\omega_C|_U \simeq \Omega^2_{\mathbb{A}^2_k/k}\Big|_U \otimes_{\mathcal{O}_U} (\mathcal{I}/\mathcal{I}^2)^\vee.$$

A global generator τ is represented on U by $(d\xi \wedge d\eta) \otimes f^\vee$, where $f^\vee(f \bmod \mathcal{I}^2) = 1$. The action sends ξ to $\lambda\xi$ and fixes η, f, and $f^\vee$, so

$$\lambda^*\tau = (\lambda\, d\xi \wedge d\eta) \otimes f^\vee = \lambda\tau.$$

Choose an elliptic curve E over k together with an embedding $\mu_4 \hookrightarrow E$ (for example, $E\colon y^2 + xy = x^3 + 1$). Let μ_4 act on E by translation and set

$$X := (C \times_k E)/\mu_4.$$

We have $H^0(X, \mathcal{O}_X) = k$. The quotient map is a μ_4-torsor, X is projective, and Lemma 7.2(1) gives regularity.

Put $D := E/\mu_4$. Choose a nonempty affine open $\operatorname{Spec} A \subset D$ and a unit $u \in A^\times$ such that the restriction of the torsor $E \to D$ has the form

$$E_A := E \times_D \operatorname{Spec} A \simeq \operatorname{Spec}\bigl(A[r]/(r^4 - u)\bigr) \longrightarrow \operatorname{Spec} A.$$

In this presentation, μ_4 fixes A and acts by $\lambda^* r = \lambda r$. Consider the diagonal μ_4-action on $U \times_k E_A$, where $U = C \cap \{z \neq 0\}$. Since r is invertible, $w := \xi/r$ is an invariant function, and the invariant subring of the coordinate ring of $U \times_k E_A$ is

$$R_0 := \bigl(A[\xi, \eta, r]/(\eta^2 - a\xi^4 - b,\ r^4 - u)\bigr)^{\mu_4} \simeq A[w, \eta]/(\eta^2 - auw^4 - b).$$

Thus $\operatorname{Spec} R_0$ is an affine open subset of X. Choose $\alpha, \beta \in \overline{k}$ with $\alpha^2 = a$ and $\beta^2 = b$. After base change to $\overline{k}$, this equation becomes

$$(\eta + \beta)^2 = \alpha^2 u w^4.$$

Put $v := \eta + \beta$. Since $E_{\overline{k}}$ is reduced, $u \in K(D_{\overline{k}})$ is not a square, and thus $R_0 \otimes_k \overline{k}$ is an integral domain. At the height-one prime $\mathfrak{p} := (w, v)$ of $R_0 \otimes_k \overline{k}$, the local ring is

$$L := (R_0 \otimes_k \overline{k})_{\mathfrak{p}} \simeq K(D_{\overline{k}})[w, v]_{(w,v)}/(v^2 - \alpha^2 u w^4).$$

The ring L has dimension one and maximal ideal $\mathfrak{m} = (w, v)L$. Since the defining equation has no linear terms, the classes of w and v form a $K(D_{\overline{k}})$-basis of $\mathfrak{m}/\mathfrak{m}^2$. Thus L has embedding dimension two and is not regular, so X is geometrically non-normal.

If σ is an invariant differential on E, then $\tau \otimes \sigma$ generates $\omega_{C \times_k E}$ and satisfies

$$\lambda^*(\tau \otimes \sigma) = \lambda(\tau \otimes \sigma).$$

Thus the character is $\chi\colon \mu_4 \to \mathbb{G}_m$, given by $\chi(\lambda) = \lambda$. This character has exact order four. By Lemma 7.1, we have $\operatorname{ord}(K_X) = 4$, so X satisfies the hypotheses of Case (B).

Example 7.5 ($p = 2$ and $\operatorname{ord}(K_X) = 3$)**.** Put $k := \mathbf{F}_2(s)$, and let

$$C := \operatorname{Proj} k[x, y, z]/(y^2 z - x^3 - sz^3).$$

The curve C is a regular plane cubic with $H^0(C, \mathcal{O}_C) = k$ and $\omega_C \simeq \mathcal{O}_C$. Let $G := \mu_3$ act on C by

$$\lambda \cdot [x : y : z] = [\lambda x : y : z].$$

Choose an elliptic curve E over k with an embedding $\mu_3 \hookrightarrow E$ (for example, $E\colon y^2 + y = x^3$), and let G act on E by translation. Set

$$X := (C \times_k E)/G.$$

As in Example 7.4, the quotient X is projective with $H^0(X, \mathcal{O}_X) = k$, and the projection induces a nonconstant morphism $X \to E/G$. By Lemma 7.2(1), X is regular.

Choose $\beta \in \overline{k}$ with $\beta^2 = s$. Then $C_{\overline{k}}$ has affine equation $(y + \beta)^2 = x^3$, so it is reduced with a unique cusp at $c = (0, \beta)$. This cusp is fixed scheme-theoretically by $G_{\overline{k}}$, so X is geometrically non-normal by Lemma 7.2(2).

By the same adjunction calculation as in Example 7.4, the character on $H^0(C, \omega_C)$ is $\lambda \mapsto \lambda$, of exact order three. Tensoring with an invariant differential on E preserves this character, so by Lemma 7.1 we have $\operatorname{ord}(K_X) = 3$.

Example 7.6 ($p = 3$ and $\operatorname{ord}(K_X) = 2$)**.** Let $k := \mathbf{F}_3(s)$ and let

$$C := \operatorname{Proj} k[x, y, z]/(y^2z - x^3 - sz^3).$$

The curve C is a regular plane cubic with $H^0(C, \mathcal{O}_C) = k$ and $\omega_C \simeq \mathcal{O}_C$. Let $G := \mu_2$ act on C by

$$\lambda \cdot [x : y : z] = [x : \lambda y : z].$$

Choose an elliptic curve E over k with an embedding $\mu_2 \hookrightarrow E$, and let G act on E by translation. Set

$$X := (C \times_k E)/G.$$

As in Example 7.4, the quotient X is projective with $H^0(X, \mathcal{O}_X) = k$. By Lemma 7.2(1), X is regular.

If $\alpha^3 = s$ in $\overline{k}$, then $C_{\overline{k}}$ has affine equation $y^2 = (x + \alpha)^3$, so it is reduced with a unique cusp at $c = (-\alpha, 0)$. This cusp is fixed scheme-theoretically by $G_{\overline{k}}$, so X is geometrically non-normal by Lemma 7.2(2).

By the same adjunction calculation as in Example 7.4, the character on $H^0(C, \omega_C)$ is $\lambda \mapsto \lambda$, of exact order two. Tensoring with an invariant differential on E preserves this character, so by Lemma 7.1 we have $\operatorname{ord}(K_X) = 2$. Thus X satisfies the hypotheses of Case (B).

Example 7.7 ($p = 3$ and $\operatorname{ord}(K_X) = 3$)**.** Let $k := \mathbf{F}_3(s, t)$ and let

$$C := \operatorname{Proj} k[x, y, z]/(sx^3 + ty^3 + z^3).$$

The curve C is a regular integral plane cubic with $H^0(C, \mathcal{O}_C) = k$ and $\omega_C \simeq \mathcal{O}_C$. Let $G := \mu_3$ act on C by

$$\lambda \cdot [x : y : z] = [\lambda x : y : z].$$

Choose an elliptic curve E over k with an embedding $\mu_3 \hookrightarrow E$, and let G act on E by translation. Set

$$X := (C \times_k E)/G.$$

As in Example 7.4, the quotient X is projective with $H^0(X, \mathcal{O}_X) = k$. By Lemma 7.2(1), X is regular.

Put $D := E/G$. As in Example 7.4, cover D by nonempty affine opens $\operatorname{Spec} A$ on which the torsor has the form

$$E_A := E \times_D \operatorname{Spec} A \simeq \operatorname{Spec}\big(A[r]/(r^3 - u)\big), \qquad u \in A^\times.$$

The action fixes A and satisfies $\lambda^* r = \lambda r$. For $U := C \cap \{z \neq 0\}$, put $v := y/z$ and $w := x/(rz)$ on $U \times_k E_A$. The invariant subring of the coordinate ring of $U \times_k E_A$ under the diagonal G-action is

$$R_0 := A[v, w]/(suw^3 + tv^3 + 1).$$

Similarly, on $V := C \cap \{y \neq 0\}$, the coordinates $v := z/y$ and $w := x/(ry)$ give the invariant ring $R_1 := A[v, w]/(suw^3 + t + v^3)$ for $V \times_k E_A$. The spectra of R_0 and R_1 cover $X \times_D \operatorname{Spec} A$. Choose $\alpha, \beta \in \overline{k}$ with $\alpha^3 = s$ and $\beta^3 = t$. After base change to $\overline{k}$, the defining equations of R_0 and R_1 become, respectively,

$$\alpha^3 uw^3 + (\beta v + 1)^3 = 0, \qquad \alpha^3 uw^3 + (v + \beta)^3 = 0.$$

Since $E_{\overline{k}}$ is integral, $u \notin K(D_{\overline{k}})^3$, so both $R_0 \otimes_k \overline{k}$ and $R_1 \otimes_k \overline{k}$ are integral domains. Hence X is geometrically reduced. As in Example 7.4, the one-dimensional local ring $(R_0 \otimes_k \overline{k})_{(w, \beta v + 1)}$ has embedding dimension two, since its defining equation has no linear terms. Thus X is geometrically non-normal.

By the same adjunction calculation as in Example 7.4, the character on $H^0(C, \omega_C)$ is $\lambda \mapsto \lambda$, of exact order three. Tensoring with an invariant differential on E preserves this character, so by Lemma 7.1 we have $\operatorname{ord}(K_X) = 3$.

7.2 A foliation quotient in characteristic five

The following example constructs a geometrically non-normal regular projective surface Y in characteristic 5 with $\operatorname{ord}(K_Y) = 5$.

Example 7.8. We first recall the surface S constructed in [WZ26, §6.1]. Let

$$\ell = \mathbf{F}_5(s_1, s_2), \qquad k = \ell(c_1, c_2), \qquad c_i^5 = s_i.$$

The construction starts with a regular projective toric k-surface $\widetilde{X}$ carrying a rank-one foliation $\mathcal{D} \subset T_{\widetilde{X}/\ell}$. Let $\sigma\colon \widetilde{X} \to X$ be the contraction of two toric curves E_1 and E_2. Then X is a normal projective toric

surface with a unique cyclic quotient singularity P of type $\frac{1}{5}(1,2)$, and $\mathcal{D}$ descends to a foliation $\mathcal{F}$ on X. Let $\pi\colon X \to S := X/\mathcal{F}$ be the quotient and put $s_0 := \pi(P)$. By [WZ26, Theorem 6.1], S is a regular projective surface, geometrically non-reduced over ℓ, with

$$H^0(S, \mathcal{O}_S) = \ell, \qquad K_S \equiv 0.$$

We will show that $\omega_S \simeq \mathcal{O}_S$ and construct a nowhere-vanishing ℓ-derivation ϑ on S satisfying $\vartheta^{[5]} = \vartheta$. The associated free μ_5-action defines a quotient $\rho\colon S \to Y := S/\vartheta$. We will prove that Y is regular, projective, and geometrically non-normal over ℓ, with $H^0(Y, \mathcal{O}_Y) = \ell$ and K_Y of exact order 5. The construction is summarized by the diagram

$$\begin{array}{ccccc} (\widetilde{X}, \mathcal{D}) & & & & \\ \Big\downarrow{\scriptstyle \sigma}\ \text{birational contraction of foliated pairs} & & & & \\ (X, \mathcal{F}) & \xrightarrow{\ \pi\ } & S := X/\mathcal{F} & \xrightarrow{\ \rho\ } & Y := S/\vartheta. \end{array}$$

Step 1: Triviality of ω_S. On the affine chart $V_1 = \operatorname{Spec} k[t,w] \subset \widetilde{X}$, the restriction $\sigma|_{V_1}\colon V_1 \to \sigma(V_1)$ is an isomorphism onto an open subset of X, so we also write V_1 for its image in X. In the function field $K = k(t,w)$, the foliation is

$$\mathcal{D} = T_{\widetilde{X}/\ell} \cap K\delta, \qquad \text{where } \delta = \partial_{c_1} + t\,\partial_{c_2} + w\,\partial_t \in \operatorname{Der}_\ell(K).$$

Put $L := \ker\delta$, so $[K : L] = 5$. Set $R := \ker(\delta|_{k[t,w]}) \subset k[t,w]$ to be the coordinate ring of $\pi(V_1)$. The functions

$$A := t - c_1 w, \qquad B := c_2 - c_1 t + \tfrac{1}{2}c_1^2 w, \qquad W := w$$

are δ-invariant. Since $k[t,w] = \bigoplus_{i=0}^4 \ell[A,B,W]c_1^i$ and $\delta(c_1) = 1$, the functions A, B, W generate R as an ℓ-algebra, with presentation

$$R \simeq \ell[A,B,W]/(F), \qquad F = B^5 - s_2 + s_1 A^5 + \tfrac{1}{2}s_1^2 W^5.$$

Adjunction for $\operatorname{Spec} R \subset \mathbb{A}^3_\ell$ gives a generator

$$\eta_0 := (dA \wedge dB \wedge dW) \otimes F^\vee$$

of ω_S on $\pi(V_1)$.

The boundary $\widetilde{X} \setminus V_1$ consists of E_1, E_2, G_1, G_2. Writing $\Gamma_i := \pi(\sigma(G_i))$, we therefore have $\operatorname{div}(\eta_0) = a\Gamma_1 + b\Gamma_2$ for some $a, b \in \mathbb{Z}$. To prove that η_0 extends to a global generator of ω_S, it remains to show that $a = b = 0$. Put $H_i := \sigma(G_i)$. We use the notation $(D \cdot C)_k$ for Mumford's numerical intersection pairing over k. The toric intersection numbers in $\widetilde{X}$ are

$$(E_1 \cdot E_1)_k = -3, \quad (E_2 \cdot E_2)_k = -2, \quad (G_1 \cdot G_1)_k = -2, \quad (G_2 \cdot G_2)_k = -1,$$
$$(E_1 \cdot E_2)_k = (E_2 \cdot G_2)_k = (G_2 \cdot G_1)_k = 1,$$

and all other intersections among these four curves are zero. The pullbacks are

$$\sigma^* H_1 = G_1, \qquad \sigma^* H_2 = G_2 + \tfrac{1}{5}E_1 + \tfrac{3}{5}E_2.$$

Therefore

$$(H_1 \cdot H_1)_k = -2, \qquad (H_1 \cdot H_2)_k = 1, \qquad (H_2 \cdot H_2)_k = -1 + \tfrac{3}{5} = -\tfrac{2}{5}.$$

Let $e_i > 0$ be defined by $\pi^*\Gamma_i = e_i H_i$. Explicitly, $\pi^* \operatorname{div}(\eta_0) = ae_1 H_1 + be_2 H_2$. Since $\operatorname{div}(\eta_0) \equiv K_S \equiv 0$, this divisor is numerically trivial. Since the intersection matrix $\big((H_i \cdot H_j)_k\big)_{1 \le i,j \le 2}$ is invertible, we obtain $a = b = 0$.

Step 2: A free μ_5-action on S. Define $\vartheta \in \operatorname{Der}_\ell(K)$ by

$$\vartheta(c_1) = -c_1, \quad \vartheta(c_2) = c_2, \quad \vartheta(t) = 2t, \quad \vartheta(w) = 3w.$$

This satisfies $\vartheta^{[5]} = \vartheta$ and $[\vartheta, \delta] = \delta$. Moreover, ϑ is diagonal on monomials in t, w, so it preserves $\mathcal{O}_X$ and L. Here $\vartheta(L) \subseteq L$ follows from $[\vartheta, \delta] = \delta$: if $f \in L$, then

$$\delta(\vartheta f) = \vartheta(\delta f) - [\vartheta, \delta](f) = 0.$$

It therefore induces a global derivation $\vartheta \in H^0(S, T_{S/\ell})$, again denoted by ϑ, with $\vartheta^{[5]} = \vartheta$. We claim that ϑ vanishes nowhere on S. For this, recall the six affine charts of $\widetilde{X}$ and the corresponding local generators ∂ of $\mathcal{D}$:

chart	(y_1, y_2)	∂	$\partial(c_1)$	$\partial(c_2)$	$\partial(y_1)$	$\partial(y_2)$
V_1	(t, w)	δ	1	t	w	0
V_2	(t, v)	$v\delta$	v	tv	1	0
V_3	$(t', \tilde{v})$	$t'^2\tilde{v}\,\delta$	$t'^2\tilde{v}$	$t'\tilde{v}$	$-t'$	$3\tilde{v}$
V_4	(t', η)	$t'\delta$	t'	1	$-t'\eta$	$3\eta^2$
V_5	$(\tilde{w}, \rho)$	$\tilde{w}^2\rho^2\delta$	$\tilde{w}^2\rho^2$	ρ	2	0
V_6	(ξ, τ)	$\xi^2\tau\,\delta$	$\xi^2\tau$	1	2	0

where $t' = 1/t$, $v = 1/w$, $\tilde{w} = w/t^3$, $\tilde{v} = 1/\tilde{w}$, $\eta = w/t^2$, $\xi = t^2/w$, and $\rho = t^5/w^2 = 1/\tau$.

We check the vanishing of ϑ separately at points $s := \pi(x)$ for $x \in X \setminus \{P\}$, and at $s_0 = \pi(P)$. For $x \in X \setminus \{P\}$, by the local structure in [WZ26], $\mathcal{D}$ is a subbundle at x and $\mathcal{O}_{X,x} = \bigoplus_{i=0}^{4} \mathcal{O}_{S,s} g^i$ for a local generator ∂ of $\mathcal{D}$ normalized by $\partial(g) = 1$. If ϑ vanished at s, then $\vartheta(a_i) \in \mathfrak{m}_s$ for every $a_i \in \mathcal{O}_{S,s}$. Every $f \in \mathcal{O}_{X,x}$ can be expressed as $f = \sum_i a_i g^i$. Using $\partial(g^i) = i g^{i-1}$, one sees that $\vartheta - \vartheta(g)\partial$ would map $\mathcal{O}_{X,x}$ into $\mathfrak{m}_x$, so

$$\vartheta \equiv \vartheta(g)\,\partial \pmod{\mathfrak{m}_x T_{\widetilde{X}/\ell}}.$$

This relation does not hold in any of the six charts in the above table. For example, on V_1, the congruences $-c_1 \equiv \vartheta(g)$, $c_2 \equiv \vartheta(g)\,t$, $2t \equiv \vartheta(g)\,w$, and $3w \equiv 0$ force $w(x) = 0$, then $t(x) = 0$, then $c_2(x) = 0$, which is absurd. Thus ϑ does not vanish at s. At s_0, recall from [WZ26] that,

$$\mathcal{O}_{X,P} = k\big[U^a T^b :\ 0 \le 2b \le 5a\big]_{\mathfrak{m}}, \qquad \partial = T\partial_T + U(T\partial_{c_1} + T^2\partial_{c_2}),$$

where $U = 1/w$, $T = t$, and $\mathfrak{m}$ is the maximal ideal of the point P. The invariant ring $\mathcal{O}_{S,s_0} = \ker(\partial|_{\mathcal{O}_{X,P}})$ has residue field k. The element $q := c_1 - t/w \in \mathcal{O}_{S,s_0}$ is a unit with residue c_1 and satisfies $\vartheta(q) = -q$, so ϑ does not vanish at s_0 either.

Since $\vartheta^{[5]} = \vartheta$ and ϑ vanishes nowhere, the associated μ_5-action on S is free.

Step 3: The quotient Y and its canonical order. Let $\rho\colon S \to Y := S/\vartheta$ be the quotient. Then ρ is a finite flat μ_5-torsor and Y is regular and projective. On $\pi(V_1) = \operatorname{Spec} R$, the derivation is diagonal:

$$\vartheta(A) = 2A, \qquad \vartheta(B) = B, \qquad \vartheta(W) = 3W.$$

Let T be an ℓ-scheme and let $\lambda \in \mu_5(T)$. The associated μ_5-action is characterized on R_T by

$$\lambda^* A = \lambda^2 A, \qquad \lambda^* B = \lambda B, \qquad \lambda^* W = \lambda^3 W,$$

corresponding to the displayed formulas for ϑ. Using the formula $\eta_0 = (dA \wedge dB \wedge dW) \otimes F^\vee$ from Step 1 and the invariance of F, we obtain

$$\lambda^* \eta_0 = \lambda^{2+1+3} \eta_0 = \lambda \eta_0$$

for every such T and λ. By Lemma 7.1, we have $\operatorname{ord}(K_Y) = 5$.

This gives a projective regular surface Y over ℓ with $H^0(Y, \mathcal{O}_Y) = \ell$, $K_Y \equiv 0$, and $\operatorname{ord}(K_Y) = 5$. Moreover, Y is geometrically non-normal and its Albanese variety is trivial: otherwise Theorem 1.1 (A) or (B) would give $12K_Y \sim 0$, contradicting $\operatorname{ord}(K_Y) = 5$. Hence Y falls in case (C), and the statement of Theorem 1.1 (C) is sharp for $p = 5$.

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